Syllabus of

Categorical dualities in logic

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Marco Abbadini

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Introduction

John Baez and James Dolan [Baez and Dolan, 2001] remarked that

> "an equation is only interesting or useful to the extent that the two sides are different."

For instance, compare

$$2 = 2 \qquad \text{with} \qquad \sum_{n=0}^{\infty} \frac{1}{2^n} = 2.$$

The first equality is correct but uninformative: both sides express the same object in the same language. The second one is interesting precisely because it connects two apparently different descriptions, and allows us to switch freely between them depending on which is more convenient for a given computation.

In this course, we will study an analogous phenomenon, but at the level of *mathematical structures*. On one side, we will have algebraic structures that arise naturally in logic — most notably *Boolean algebras*, which provide an algebraic semantics for classical propositional logic. On the other side, we will encounter structures of a completely different nature — in this case, *Stone spaces*, which are certain topological spaces.

The connection between these two worlds is not a literal equality, but a *categorical duality* (a.k.a. categorical dual equivalence): a two-way translation that preserves all information. This means that we can translate a problem about Boolean algebras into a corresponding problem about Stone spaces, and vice versa. In practice, this is useful because, on the side of Stone spaces, many constructions are simpler, and one can use geometric intuition.

A slogan to keep in mind is:

> *Categorical dualities in logic relate "algebras of formulas" to "spaces of models".*

We begin with the simplest case: classical propositional logic, which is modeled by Boolean algebras and Stone spaces (Stone, 1936).

Later, we will see an analogous duality for classical *first-order* logic, where quantifiers enter the picture.

Chapter 1

Classical propositional logic

In the first part of the course, we will see *Stone duality*: a connection (in the form of a categorical duality) between *Boolean algebras* and *Stone spaces*. Informally, Boolean algebras encode the *syntax* of classical propositional logic (algebras of formulas), while Stone spaces encode its *semantics* (spaces of models).

1.1 From classical propositional logic towards Boolean algebras

1.1.1 Syntax: propositional languages, formulas

Definition 1.1.1 (Propositional language, propositional symbols)**.** A ***(propositional) language*** $\mathcal{L}$ is a set; its elements are called ***propositional symbols*** (or also *propositional variables*).

We use the fancy name "(propositional) language" for a plain set just to declare the usage we want to make out of it.

Propositional symbols are typically denoted by $p, q, r, \ldots$.

The connectives of classical propositional logic are

$$\vee, \quad \wedge, \quad \neg, \quad 0, \quad 1,$$

where $\vee$ is a binary operation denoting *or* (sometimes called *join*), $\wedge$ is a binary operation denoting *and* (sometimes called *meet*), $\neg$ is a unary operation denoting *not* (i.e. negation, or complement), 0 is a constant symbol denoting *false* (bottom) and 1 is a constant symbol denoting *true* (top).

Definition 1.1.2 (Formula)**.** The set of ***formulas*** $\mathrm{Form}(\mathcal{L})$ is defined inductively as follows:

- every propositional symbol $p \in \mathcal{L}$ is a formula;
- if $\varphi, \psi \in \mathrm{Form}(\mathcal{L})$, then $(\varphi \vee \psi) \in \mathrm{Form}(\mathcal{L})$ and $(\varphi \wedge \psi) \in \mathrm{Form}(\mathcal{L})$;
- if $\varphi \in \mathrm{Form}(\mathcal{L})$, then $(\neg\varphi) \in \mathrm{Form}(\mathcal{L})$;
- 0 and 1 are formulas.

In other words, $\mathrm{Form}(\mathcal{L})$ is the smallest set containing $\mathcal{L}$ and closed under $\vee$, $\wedge$, $\neg$, 0 and 1.

We use the standard abbreviations for implications and bi-implication:

$$\varphi \to \psi \;\coloneqq\; \neg\varphi \vee \psi, \qquad \varphi \leftrightarrow \psi \;\coloneqq\; (\varphi \to \psi) \wedge (\psi \to \varphi).$$

While propositional symbols are usually denoted by $p, q, r, \ldots$, formulas are typically denoted by $\varphi, \psi, \sigma, \ldots$.

1.1.2 Semantics: valuations, models, semantic equivalence

Valuations

Set $2 := \{0, 1\}$ (the "set of truth values").

Definition 1.1.3 (Valuation, interpretation of formulas)**.** A ***valuation*** on $\mathcal{L}$ is a function $v\colon \mathcal{L} \to 2$. Given a valuation v, its unique extension[a] to all formulas is a map

$$\bar{v}\colon \mathrm{Form}(\mathcal{L}) \longrightarrow 2$$

defined by recursion on the complexity of formulas:

$$\bar{v}(p) = v(p) \quad (p \in \mathcal{L}), \qquad \bar{v}(0) = 0, \qquad \bar{v}(1) = 1,$$

$$\bar{v}(\neg\varphi) = \neg\,\bar{v}(\varphi), \qquad \bar{v}(\varphi \vee \psi) = \bar{v}(\varphi) \vee \bar{v}(\psi), \qquad \bar{v}(\varphi \wedge \psi) = \bar{v}(\varphi) \wedge \bar{v}(\psi),$$

where on the right-hand side we use the usual Boolean operations on $2 = \{0, 1\}$, which you can find in Remark 1.1.4 below.

[a] This is a special case of a standard universal-algebraic fact: for any algebraic signature Σ, the forgetful functor $\mathsf{Alg}(\Sigma) \to \mathsf{Set}$ has a left adjoint sending a set X to the *free* Σ-*algebra* on X, whose underlying set can be described as the set of Σ-terms over X. For the signature $\{\vee, \wedge, \neg, 0, 1\}$, the left adjoint maps a set $\mathcal{L}$ to $\mathrm{Form}(\mathcal{L})$, the unit at $\mathcal{L}$ is the inclusion $\mathcal{L} \hookrightarrow \mathrm{Form}(\mathcal{L})$, and the extension $v \mapsto \bar{v}$ is given by the universal property of the unit.

Remark 1.1.4 (Boolean operations on $2 = \{0, 1\}$)**.** By convention,

$$0 \wedge 0 = 0, \;\; 0 \wedge 1 = 0, \;\; 1 \wedge 0 = 0, \;\; 1 \wedge 1 = 1,$$

$$0 \vee 0 = 0, \;\; 0 \vee 1 = 1, \;\; 1 \vee 0 = 1, \;\; 1 \vee 1 = 1,$$

$$\neg 0 = 1, \;\; \neg 1 = 0.$$

Semantic equivalence

Definition 1.1.5 (Satisfaction: "valuation $\models$ formula")**.** Let $\mathcal{L}$ be a propositional language. Given a valuation $v\colon \mathcal{L} \to 2$ and a formula $\varphi \in \mathrm{Form}(\mathcal{L})$, we write

$$v \models \varphi$$

and we say that v ***satisfies*** φ if

$$\bar{v}(\varphi) = 1.$$

Definition 1.1.6 (Semantic equivalence)**.** Let $\varphi, \psi \in \mathrm{Form}(\mathcal{L})$. We write $\varphi \equiv \psi$ and say that φ and ψ are ***semantically equivalent***[a] if, for every valuation $v\colon \mathcal{L} \to 2$,

$$v \models \varphi \iff v \models \psi$$

(i.e., $\bar{v}(\varphi) = \bar{v}(\psi)$).

[a] In Definition 1.1.6, "semantic" is in opposition to "syntactic": two formulas are *syntactically equivalent* if they are interprovable in a certain proof system, which we do not have the time to see here. Let me just mention that Stone's Representation Theorem, which will be seen later and which is the core of this chapter, can be seen as an algebraic way to affirm that the syntactic and semantic notions of equivalence coincide.

In other words, φ and ψ are equivalent if and only if they have the same truth table (a function from $\mathcal{L}$ to 2 corresponds to a row of a truth table).

Example 1.1.7. Let $\mathcal{L} = \{ p, q \}$. Then $p \vee q \equiv q \vee p$, as can be checked via the truth table

p	q	$p \vee q$	$q \vee p$
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

Example 1.1.8. If $\mathcal{L} = \{ p \}$ has one variable, then every formula is equivalent to exactly one of

$$0, \quad p, \quad \neg p, \quad 1.$$

(For instance, $\neg\neg p \equiv p$, $p \vee \neg p \equiv 1$, and $p \wedge \neg p \equiv 0$.) Thus $\mathrm{Form}(\mathcal{L})/\equiv$ has 4 elements.

One may prove that, in a language $\mathcal{L} = \{ p, q \}$ with two propositional symbols, there are 16 equivalence classes of formulas.[1]

Adding assumptions: theories and semantic equivalence modulo a theory

We can incorporate semantic assumptions by restricting the class of admissible valuations.

Definition 1.1.9 (Propositional theory)**.** A ***(propositional) theory*** $\mathcal{T}$ in a propositional language $\mathcal{L}$ is a subset $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$.

Definition 1.1.10 (Model)**.** Let $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ be a propositional theory in a propositional language. A ***model*** of $\mathcal{T}$ is a function $v \colon \mathcal{L} \to 2$ that satisfies every element of $\mathcal{T}$. We denote by $\mathrm{Mod}(\mathcal{T})$ the set of models of $\mathcal{T}$.

Example 1.1.11. Let $\mathcal{L} = \{ p, q \}$ and let $\mathcal{T} = \{ p \vee q \}$. Then $\mathrm{Mod}(\mathcal{T})$ consists of the three valuations

$$(p, q) = (1, 0),\ (1, 1),\ (0, 1).$$

Definition 1.1.12 (Semantic equivalence modulo a theory)**.** Let $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ be a propositional theory and let $\varphi, \psi \in \mathrm{Form}(\mathcal{L})$. We say that φ and ψ are ***semantically equivalent modulo the theory*** $\mathcal{T}$ (or ***relative to*** $\mathcal{T}$, or ***semantically*** $\mathcal{T}$***-equivalent***), and write $\varphi \equiv_{\mathcal{T}} \psi$, if for every $v \in \mathrm{Mod}(\mathcal{T})$ we have

$$v \models \varphi \iff v \models \psi$$

Example 1.1.13. Let $\mathcal{L} = \{ p, q \}$ and let $\mathcal{T} = \{ p \vee q \}$. One can check that $p \vee q \equiv_{\mathcal{T}} 1$, $\neg p \wedge \neg q \equiv_{\mathcal{T}} 0$ and $p \vee \neg q \equiv_{\mathcal{T}} p$.

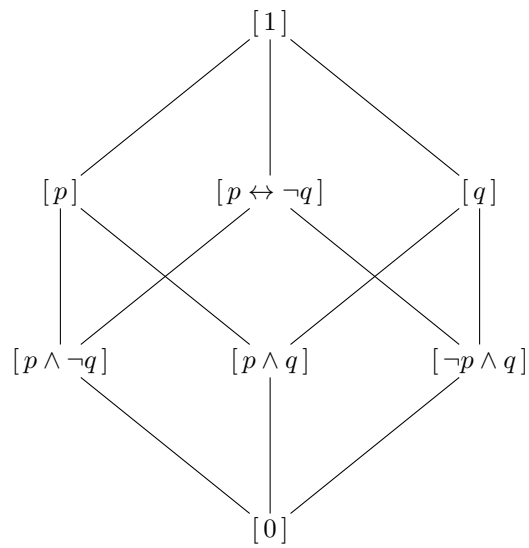
The relation $\equiv_{\mathcal{T}}$ is again an equivalence relation, and we write $\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$ for the corresponding quotient set. It is convenient to partially order equivalence classes by *implication*:

$$\varphi \leq_{\mathcal{T}} \psi \quad :\iff \quad \forall v \in \mathrm{Mod}(\mathcal{T}),\ \big((v \models \varphi) \Rightarrow (v \models \psi)\big).$$

(So "$\varphi \leq_{\mathcal{T}} \psi$" means that φ *implies* ψ on every model of $\mathcal{T}$.)

Example 1.1.14. Let $\mathcal{L} = \{ p, q \}$ and $\mathcal{T} = \{ p \vee q \}$. Then $\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$ has 8 elements and its Hasse diagram (with respect to $\leq_{\mathcal{T}}$) can be drawn as follows:

[1] More generally, if $\mathcal{L}$ is finite of cardinality n, then $|\mathrm{Form}(\mathcal{L})/\equiv| = 2^{2^n}$. Indeed, every formula determines a truth function $2^{\mathcal{L}} \to 2$, and vice versa one can prove that every function $2^{\mathcal{L}} \to 2$ is the truth function of a formula (this is called "functional completeness"), so that $|\mathrm{Form}(\mathcal{L})/\equiv| = 2^{2^{|\mathcal{L}|}}$. In particular, for $\mathcal{L} = \{ p, q \}$ we have $|\mathrm{Form}(\mathcal{L})/\equiv| = 2^{2^2} = 16$.



Boolean algebras (whose definition we will see soon) are meant to capture the algebraic structures of the form

$$\langle \mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}; \vee, \wedge, \neg, 0, 1 \rangle,$$

for $\mathcal{L}$ a propositional language and $\mathcal{T}$ a propositional theory in $\mathcal{L}$; here, $\vee, \wedge, \neg, 0, 1$ are defined on $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ by setting

$$[\,\varphi\,] \vee [\,\psi\,] \;\coloneqq\; [\,\varphi \vee \psi\,], \qquad [\,\varphi\,] \wedge [\,\psi\,] \;\coloneqq\; [\,\varphi \wedge \psi\,], \qquad \neg\,[\,\varphi\,] \;\coloneqq\; [\,\neg\varphi\,], \qquad 0 \;\coloneqq\; [\,0\,], \qquad 1 \;\coloneqq\; [\,1\,].$$

(These are well-defined operations.)

In other words, Boolean algebras are meant to capture the algebras of formulas modulo a theory.

1.2 Boolean algebras

The quotient posets in Example 1.1.14 already display a key phenomenon: the logical connectives $\wedge$ and $\vee$ behave like "infimum" and "supremum" with respect to the implication order.

To recall the definition of infimum and supremum, let P be a poset. For $x, y \in P$, an *infimum* (or *greatest lower bound*) of $\{\,x, y\,\}$ is an element $x \wedge y \in P$ such that

$$x \wedge y \leq x, \qquad x \wedge y \leq y,$$

and, for every $z \in P$, if $z \leq x$ and $z \leq y$ then $z \leq x \wedge y$. Similarly, a *supremum* (or *least upper bound*) of $\{\,x, y\,\}$ is an element $x \vee y \in P$ such that

$$x \leq x \vee y, \qquad y \leq x \vee y,$$

and, for every $z \in P$, if $x \leq z$ and $y \leq z$ then $x \vee y \leq z$.

More generally, for any subset S of a poset P, an ***infimum*** of S (relative to P) is a greatest lower bound of S in P and a ***supremum*** of S (relative to P) is a smallest upper bound of S in P. Infima and suprema, if they exist, are unique. Note that being the infimum of $\varnothing$ relative to P means being the maximum of P, and being the supremum of $\varnothing$ relative to P means being the minimum of P.

One can show that, in the poset $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ ordered by implication, $[\,\varphi \wedge \psi\,]$ is the infimum of $[\,\varphi\,]$ and $[\,\psi\,]$, and $[\,\varphi \vee \psi\,]$ is the supremum of $[\,\varphi\,]$ and $[\,\psi\,]$.[2]

[2] Proof: We prove the statement for $\wedge$; the case of $\vee$ is analogous. For every model $v \in \mathrm{Mod}(\mathcal{T})$, if $\bar{v}(\varphi \wedge \psi) = 1$ then $\bar{v}(\varphi) = 1$ and $\bar{v}(\psi) = 1$, hence $[\,\varphi \wedge \psi\,] \leq_{\mathcal{T}} [\,\varphi\,]$ and $[\,\varphi \wedge \psi\,] \leq_{\mathcal{T}} [\,\psi\,]$. Now let $[\,\rho\,]$ be any lower bound of $[\,\varphi\,]$ and $[\,\psi\,]$. This means that, for every $v \in \mathrm{Mod}(\mathcal{T})$, $\bar{v}(\rho) = 1$ implies both $\bar{v}(\varphi) = 1$ and $\bar{v}(\psi) = 1$. Therefore $\bar{v}(\rho) = 1$ implies $\bar{v}(\varphi \wedge \psi) = 1$, i.e. $[\,\rho\,] \leq_{\mathcal{T}} [\,\varphi \wedge \psi\,]$.

1.2.1 Lattices

Order-theoretic definition of lattices

Definition 1.2.1 (Lattice: order theoretic presentation)**.** A ***lattice*** is a poset $(L, \leq)$ in which every pair of elements admits an infimum and a supremum.

Example 1.2.2 (Power-set lattice)**.** Let X be a set and consider the poset $(\mathcal{P}(X), \subseteq)$. Then $(\mathcal{P}(X), \subseteq)$ is a lattice, with

$$A \wedge B := A \cap B \qquad \text{and} \qquad A \vee B := A \cup B.$$

Equational definition of lattices

Lattices can also be presented *algebraically*, by taking $\wedge$ and $\vee$ as primitive operations and listing a small family of identities. This is useful because identities are stable under the kind of constructions we will use later (products, subalgebras, quotients).

Definition 1.2.3 (Lattice: equational presentation)**.** A ***lattice*** is a set L equipped with two binary operations

$$\wedge, \vee \colon L \times L \to L$$

such that:

1. (commutativity) for all $a, b \in L$, $a \wedge b = b \wedge a$ and $a \vee b = b \vee a$;
2. (associativity) for all $a, b, c \in L$, $(a \wedge b) \wedge c = a \wedge (b \wedge c)$ and $(a \vee b) \vee c = a \vee (b \vee c)$;
3. (absorption) for all $a, b \in L$, $a \vee (a \wedge b) = a$ and $a \wedge (a \vee b) = a$.

Remark 1.2.4. If L is a lattice in the equational sense, one can recover an order by setting

$$a \leq b \quad :\Longleftrightarrow \quad a \wedge b = a,$$

equivalently $a \vee b = b$. With this order, the operations $\wedge$ and $\vee$ are precisely infimum and supremum, so Definitions 1.2.1 and 1.2.3 are equivalent viewpoints.

1.2.2 Bounded lattices

Definition 1.2.5 (Bounded lattice)**.** A lattice is ***bounded*** if it has a least element 0 and a greatest element 1, i.e. elements such that $0 \leq a \leq 1$ for all a. Equivalently (in the equational presentation), 0 and 1 satisfy

1. For all a, $a \wedge 0 = 0$ (equivalently: $a \vee 0 = a$),
2. For all a, $a \vee 1 = 1$ (equivalently; $a \wedge 1 = a$).

1.2.3 Distributive lattices

Definition 1.2.6 (Distributive lattice)**.** A lattice L is ***distributive*** if it satisfies any (and hence both) of the following equivalent[a] conditions:

1. for all $a, b, c \in L$, $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$.
2. for all $a, b, c \in L$, $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$,

[a] For a proof of the equivalence between the two, see [Birkhoff, 1967, Sec. 6, Thm. 9, p. 11].

Example 1.2.7. The two-element chain is distributive:

Example 1.2.8. For every set X, the power-set lattice $\mathcal{P}(X)$ is distributive.

Two small lattices play a special role as "minimal" obstructions to distributivity. They are usually denoted by M_3 (the *diamond*) and N_5 (the *pentagon*).

M_3 $\qquad$ N_5 (1.1)

These are not distributive; one can easily verify that $a \vee (b \wedge c) \neq (a \vee b) \wedge (a \vee c)$.

In fact, one can prove that a lattice L is distributive if and only if it does not contain a sublattice isomorphic to M_3 or N_5, i.e., there is no injective map $M_3 \to L$ or $N_5 \to L$ preserving both $\wedge$ and $\vee$.[3]

1.2.4 Boolean algebras

Definition 1.2.9 (Boolean algebra)**.** A ***Boolean algebra*** is an algebraic structure

$$\langle B; \vee, \wedge, \neg, 0, 1 \rangle$$

(arities: 2, 2, 1, 0, 0, respectively) such that:

1. $\langle B; \vee, \wedge, 0, 1 \rangle$ is a bounded distributive lattice;
2. for every $a \in B$,
$$a \wedge \neg a = 0 \qquad \text{and} \qquad a \vee \neg a = 1.$$

The element $\neg a$ is called the *complement* of a.

Remark 1.2.10. All axioms in the definition of Boolean algebras are *equational*, i.e. they are identities of the form

$$\forall x_1, \ldots, x_n, \quad t_1(x_1, \ldots, x_n) = t_2(x_1, \ldots, x_n)$$

between terms.

Remark 1.2.11. A Boolean algebra is completely determined by its underlying partial order. This is because: $\vee, \wedge, 0, 1$ are the binary supremum, binary infimum, smallest element and greatest element, and, in every bounded distributive lattice L, if an element a has a complement (i.e., there is an element b such that $a \wedge b = 0$ and $a \vee b = 1$) it is unique.[4]

[3] For a textbook reference, see, e.g., [Davey and Priestley, 2002, 4.10].

[4] This is not true for arbitrary bounded lattice: for example, in the bounded lattice M_3 in (1.1), both b and c are complements of a, and in N_5 both a and c are complements of b.

Example 1.2.12. The prototypical example of a Boolean algebra is $2 = \{0, 1\}$ with partial order $0 \leq 1$. The Boolean operations are those described in Remark 1.1.4.

Proposition 1.2.13 (Form($\mathcal{L}$)/$\equiv_{\mathcal{T}}$ is a Boolean algebra)**.** *Let $\mathcal{L}$ be a propositional language and $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ a theory. Then the quotient set $\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$ becomes a Boolean algebra by setting*

$$[\varphi] \vee [\psi] := [\varphi \vee \psi], \qquad [\varphi] \wedge [\psi] := [\varphi \wedge \psi], \qquad \neg [\varphi] := [\neg\varphi], \qquad 0 := [0], \qquad 1 := [1].$$

Proof sketch. Step 1: the operations are well defined. We show this for $\vee$; the cases of $\wedge$ and $\neg$ are analogous. Suppose $\varphi \equiv_{\mathcal{T}} \varphi'$ and $\psi \equiv_{\mathcal{T}} \psi'$. Then for every $v \in \mathrm{Mod}(\mathcal{T})$ we have $\bar{v}(\varphi) = \bar{v}(\varphi')$ and $\bar{v}(\psi) = \bar{v}(\psi')$. Using the truth table for $\vee$ in $2 = \{0, 1\}$ we compute:

$$\bar{v}(\varphi \vee \psi) = \bar{v}(\varphi) \vee \bar{v}(\psi) = \bar{v}(\varphi') \vee \bar{v}(\psi') = \bar{v}(\varphi' \vee \psi').$$

Hence, $\varphi \vee \psi \equiv_{\mathcal{T}} \varphi' \vee \psi'$, and so $[\varphi] \vee [\psi]$ does not depend on the choice of representatives. The constants $0 = [0]$ and $1 = [1]$ are trivially well defined.

Step 2: the Boolean algebra identities hold. As an example, we show that the axiom $a \wedge \neg a = 0$ holds. Fix $\varphi \in \mathrm{Form}(\mathcal{L})$. For every $v \in \mathrm{Mod}(\mathcal{T})$ we compute

$$\bar{v}(\varphi \wedge \neg\varphi) = \bar{v}(\varphi) \wedge \bar{v}(\neg\varphi) = \bar{v}(\varphi) \wedge \neg\bar{v}(\varphi) = 0 = \bar{v}(0),$$

where we used the recursive definition of $\bar{v}$ and the fact that $b \wedge \neg b = 0$ for all $b \in 2$. Hence, $\varphi \wedge \neg\varphi \equiv_{\mathcal{T}} 0$, and therefore

$$[\varphi] \wedge \neg [\varphi] = [0] = 0.$$

This is the general mechanism: any identity between Boolean terms can be checked pointwise in 2 under every valuation. To give more details, let $t_1(x_1, \ldots, x_n) = t_2(x_1, \ldots, x_n)$ be any Boolean algebra identity (an equation between terms built from $\vee, \wedge, \neg, 0, 1$). To check it in $\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$, pick arbitrary formulas $\varphi_1, \ldots, \varphi_n \in \mathrm{Form}(\mathcal{L})$ and consider the two formulas $t_1(\varphi_1, \ldots, \varphi_n)$ and $t_2(\varphi_1, \ldots, \varphi_n)$. For every $v \in \mathrm{Mod}(\mathcal{T})$, the evaluation map $\bar{v}\colon \mathrm{Form}(\mathcal{L}) \to 2$ respects the connectives, hence

$$\bar{v}\big(t_i(\varphi_1, \ldots, \varphi_n)\big) = t_i\big(\bar{v}(\varphi_1), \ldots, \bar{v}(\varphi_n)\big) \qquad (i = 1, 2).$$

Since the identity $t_1 = t_2$ holds in the two-element Boolean algebra 2, the right-hand sides are equal for all $v \in \mathrm{Mod}(\mathcal{T})$, so $t_1(\varphi_1, \ldots, \varphi_n) \equiv_{\mathcal{T}} t_2(\varphi_1, \ldots, \varphi_n)$. Therefore, the induced operations on equivalence classes satisfy all Boolean algebra axioms. ■

The Boolean algebra

$$\langle \mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}; \vee, \wedge, \neg, 0, 1 \rangle$$

is called the ***Lindenbaum–Tarski algebra*** of $\mathcal{T}$.

Proposition 1.2.13 states that every Lindenbaum–Tarski algebra is a Boolean algebra. Later on (Theorem 1.4.4), we will see that also the converse holds: every Boolean algebra is isomorphic to $\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$ for some language $\mathcal{L}$ and theory $\mathcal{T}$. This will be a consequence of Stone's Representation Theorem (or of the crucial lemmas used in proving Stone's Representation Theorem). It will provide a guarantee that the definition of Boolean algebras is "the correct one".

Example 1.2.14 (Power set Boolean algebras)**.** Let X be a set. The power set $\mathcal{P}(X)$ with the inclusion order is a Boolean algebra, with operations

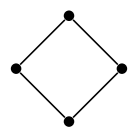
$$A \wedge B \coloneqq A \cap B, \qquad A \vee B \coloneqq A \cup B, \qquad \neg A \coloneqq X \setminus A, \qquad 0 \coloneqq \varnothing, \qquad 1 \coloneqq X.$$

See the footnote for a proof.[5]

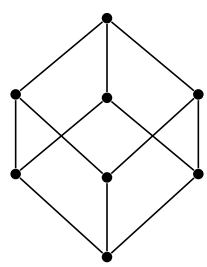
Below, we draw the Hasse diagrams of the power sets of sets of cardinality 0, 1, 2, and 3. These are the smallest Boolean algebras, and probably the only ones that one can draw without losing one's sight. (The next one has 16 elements.)

The *trivial Boolean algebra*, of cardinality 1, isomorphic to $\mathcal{P}(\varnothing)$

The Boolean algebra $2 = \{0, 1\}$, of cardinality 2, isomorphic to $\mathcal{P}(\{\, x \,\})$



A Boolean algebra of cardinality 4, isomorphic to $\mathcal{P}(\{\, x, y \,\})$



A Boolean algebra of cardinality 8, isomorphic to $\mathcal{P}(\{\, x, y, z \,\})$

There are examples of Boolean algebras that are not isomorphic to any power set. The following one is probably the simplest example.

Example 1.2.15 (The algebra of finite and cofinite subsets of $\mathbb{N}$)**.** Consider the collection

$$\mathrm{FC}(\mathbb{N}) \;\coloneqq\; \{\, A \subseteq \mathbb{N} \mid A \text{ is finite, or cofinite} \,\} \;\subseteq\; \mathcal{P}(\mathbb{N})$$

where "A *cofinite*" means that $\mathbb{N} \setminus A$ is finite. To show that this is indeed a Boolean algebra, the following is of help.

Remark 1.2.16 (Boolean subalgebras of power sets)**.** If $\mathcal{A} \subseteq \mathcal{P}(X)$ contains $\varnothing$ and X and is closed under $\cup$, $\cap$ and complement in X, then $\mathcal{A}$ is a Boolean algebra, as well.

This follows from the fact that Boolean algebras are equationally definable and hence closed under subalgebras.

[5] One way to prove that it is a Boolean algebra is by observing it is isomorphic to the power $\prod_{x \in X} 2$ of 2, that 2 is a Boolean algebra, and that equationally defined classes of algebras are closed under products (by the easy direction in Birkhoff's theorem).

Now one can prove that $\mathrm{FC}(\mathbb{N})$ is a Boolean algebra:

- $\varnothing$ and $\mathbb{N}$ belong to $\mathrm{FC}(\mathbb{N})$;
- $\mathrm{FC}(\mathbb{N})$ is closed under complement in $\mathbb{N}$: if A is finite then $\mathbb{N} \setminus A$ is cofinite, and if A is cofinite then $\mathbb{N} \setminus A$ is finite;
- $\mathrm{FC}(\mathbb{N})$ is closed under unions: if A and B are finite then $A \cup B$ is finite, while if at least one of A, B is cofinite then $A \cup B$ is cofinite.
- Analogously, $\mathrm{FC}(\mathbb{N})$ is closed under intersections.

The Boolean algebra $\mathrm{FC}(\mathbb{N})$ is infinite but much smaller than $\mathcal{P}(\mathbb{N})$: for instance, the set of even numbers is neither finite nor cofinite, hence it does not belong to $\mathrm{FC}(\mathbb{N})$. Moreover, $\mathrm{FC}(\mathbb{N})$ is *countably infinite* (i.e., in bijection with $\mathbb{N}$). This also shows that it is not isomorphic to any power set, since no power set is countably infinite: if X is finite then $\mathcal{P}(X)$ is finite, and if X is infinite then $\mathcal{P}(X)$ is uncountable.

Remark 1.2.17 (A different obstruction: completeness)**.** More generally, for any set X, one can consider the Boolean subalgebra

$$\mathrm{FC}(X) \coloneqq \{\, A \subseteq X \mid A \text{ is finite or cofinite} \,\} \subseteq \mathcal{P}(X).$$

If X is infinite, then $\mathrm{FC}(X)$ is *not complete* as a Boolean algebra: there are families that do not admit a supremum. For instance, choose a subset $Y \subsetneq X$ such that both Y and $X \setminus Y$ are infinite, and consider the family $\{\, \{\, y \,\} \mid y \in Y \,\} \subseteq \mathrm{FC}(X)$. Its union in $\mathcal{P}(X)$ is Y, which does not belong to $\mathrm{FC}(X)$. Moreover, there is no least element of $\mathrm{FC}(X)$ containing all singletons $\{\, y \,\}$ (one can always remove a point from any upper bound and still obtain a cofinite upper bound), hence the family has no supremum in $\mathrm{FC}(X)$. By contrast, every power set $\mathcal{P}(X)$ is complete (arbitrary unions exist), so $\mathrm{FC}(X)$ for X infinite cannot be isomorphic to any power set.

1.3 Stone's Representation Theorem

Remark 1.2.16 gives a zoo of examples: every *Boolean subalgebra* of some power set is a Boolean algebra. Stone's Representation Theorem states that all Boolean algebras are of this form! We state it now, and we will prove it later Theorem 1.3.20.

Theorem (Stone's Representation Theorem for Boolean algebras)**.** *For every Boolean algebra B there is a set X such that B is isomorphic to a Boolean subalgebra of the power set $\mathcal{P}(X)$.*

This means that for every Boolean algebra B there are a set X and an injective map

$$\iota \colon B \hookrightarrow \mathcal{P}(X)$$

such that, under ι, the operations on B correspond to intersection, union, complement, empty set and whole set in X:

$$\iota(a \wedge b) = \iota(a) \cap \iota(b), \qquad \iota(a \vee b) = \iota(a) \cup \iota(b), \qquad \iota(\neg a) = X \setminus \iota(a), \qquad \iota(0) = \varnothing, \qquad \iota(1) = X.$$

The next goal is to prove Stone's Representation Theorem for Boolean algebras. We will first need to present auxiliary notions and lemmas.

Given a Boolean algebra B, how can we find a set X such that B embeds into the power set $\mathcal{P}(X)$ of X, as required by the statement of Stone's Representation Theorem?

Idea, from a logical perspective: In the special Boolean algebra $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ (for $\mathcal{L}$ a language and $\mathcal{T}$ a theory), an equivalence class $[\,\varphi\,]$ can be identified with the set of models $v \in \mathrm{Mod}(\mathcal{T})$ such that $\bar{v}(\varphi) = 1$;

i.e., a formula can be identified with the models that satisfy it. This gives a very concrete embedding into a power set: the power set of models of $\mathcal{T}$. So, the idea for $B = \mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$ is to take

$$X = \mathrm{Mod}(\mathcal{T}).$$

To translate this idea to a general Boolean algebra B, we note that a model

$$v \colon \mathcal{L} \longrightarrow 2$$

of $\mathcal{T}$ induces a function

$$\bar{v} \colon \mathrm{Form}(\mathcal{L}) \longrightarrow 2$$

which passes to the quotient

$$\begin{aligned} \mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}} &\longrightarrow 2 \\ [\,\varphi\,] &\longmapsto \bar{v}(\varphi). \end{aligned}$$

In fact, one can prove that the models of $\mathcal{T}$ are in bijection with the Boolean homomorphisms (i.e., functions preserving all Boolean connectives, see Definition 1.3.1 below) from $\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$ to 2. We recall the definition of Boolean homomorphism, and then we state this bijection in Proposition 1.3.2.

Definition 1.3.1 (Boolean homomorphism)**.** Let A and B be Boolean algebras. A ***Boolean homomorphism*** (or, simply, a *homomorphism*) $f \colon A \to B$ is a function that preserves all Boolean operations, i.e., such that, for all $a, b \in A$,

$$f(a \vee b) = f(a) \vee f(b), \qquad f(a \wedge b) = f(a) \wedge f(b), \qquad f(\neg a) = \neg f(a), \qquad f(0) = 0, \qquad f(1) = 1.$$

(These conditions are redundant; for example, it is enough to ask the preservation of $\wedge, \neg, 1$.)

Proposition 1.3.2 (Models $\overset{1:1}{\leftrightarrow}$ homomorphisms to 2)**.** *Let $\mathcal{L}$ be a propositional language and $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ a propositional theory. The following function defines a bijection*

$$\begin{aligned} \mathrm{Mod}(\mathcal{T}) &\longrightarrow \hom(\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}, 2) \\ v &\longmapsto \big([\,\varphi\,] \mapsto \bar{v}(\varphi)\big). \end{aligned}$$

Proof. We skip the proof. ∎

This suggests that, for a general Boolean algebra B, we shall take

$$X = \hom(B, 2),$$

the set of homomorphisms from B to 2.

Idea, from another perspective: If $B \subseteq \mathcal{P}(X)$ is a Boolean subalgebra of $\mathcal{P}(X)$, then every element $x \in X$ induces a function

$$\begin{aligned} B &\longrightarrow 2 \\ A &\longmapsto \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{otherwise,} \end{cases} \end{aligned}$$

which is a Boolean homomorphism. This suggests a close relationship between X and $\hom(B, 2)$. This suggests, from yet another perspective, that to prove Stone's Representation Theorem we shall take

$$X = \hom(B, 2).$$

1.3.1 Ultrafilters

A homomorphism $f\colon B \to 2$ can be encoded via $f^{-1}[\{1\}]$. The subsets of B arising as $f^{-1}[\{1\}]$ for some homomorphism $f\colon B \to 2$ can be characterized as the *ultrafilters*. To define this, we first define filters (of which ultrafilters are special instances) and ideals.

Think of a filter as a coherent notion of 'truth': a set of propositions closed under strengthening and finite conjunction.

Definition 1.3.3 (Filter)**.** A ***filter*** of a Boolean algebra B is a subset $F \subseteq B$ such that:

1. F is *upward closed*, i.e., if $x \in F$ and $x \leq y$, then $y \in F$;
2. F is *closed under finite meets*, i.e.,
   (a) if $x, y \in F$, then $x \wedge y \in F$;
   (b) $1 \in F$.

A filter F is *proper* if $0 \notin F$.

Turning the definition of a filter upside down, we get the following.

Definition 1.3.4 (Ideal)**.** An ***ideal*** of a Boolean algebra B is a subset $I \subseteq B$ such that:

1. I is *downward closed*, i.e., if $x \in I$ and $y \leq x$, then $y \in I$;
2. I is *closed under finite joins*, i.e.,
   (a) if $x, y \in I$, then $x \vee y \in I$;
   (b) $0 \in I$.

An ultrafilter is what you get when you push consistency to the limit: every proposition gets a definitive yes/no answer.

Definition 1.3.5 (Ultrafilter)**.** An ***ultrafilter*** of a Boolean algebra is a filter whose complement is an ideal.

Spelling out the details, this means that an ultrafilter is a filter F such that

1. if $x \vee y \in F$, then $x \in F$ or $y \in F$;
2. $0 \notin F$.

There is also another useful characterization of ultrafilters.

Lemma 1.3.6 (Ultrafilters via negation)**.** *The ultrafilters of a Boolean algebra B are precisely the filters F such that, for every $x \in B$, exactly one between x and $\neg x$ belongs to F.*

Proof. ($\Rightarrow$). Suppose that F is an ultrafilter, and let $x \in B$. Since $1 = x \vee \neg x \in F$, we have $x \in F$ or $\neg x \in F$. Moreover, they cannot both lie in F because $x \wedge \neg x = 0 \notin F$.

($\Leftarrow$). Suppose that F is a filter such that, for every $x \in B$, exactly one between x and $\neg x$ belongs to F.

- Suppose $x \vee y \in F$, and let us prove that $x \in F$ or $y \in F$. If $x \in F$, we are done. Otherwise, from $x \notin F$ and from the hypothesis on F, we deduce $\neg x \in F$. Then, using distributivity,
$$F \ni (x \vee y) \wedge \neg x = (x \wedge \neg x) \vee (y \wedge \neg x) = y \wedge \neg x.$$
  By upward closure, this implies $y \in F$ (since $y \wedge \neg x \leq y$). $x \in F$ or $y \in F$.

- Since 1 belongs to F, its negation $\neg 1 = 0$ does not. ■

Proposition 1.3.7 (Ultrafilters $\overset{1:1}{\leftrightarrow}$ homomorphisms to 2)**.** *For every Boolean algebra B, there is a bijection*

$$\begin{aligned} \hom(B, 2) &\longleftrightarrow \{\, \textit{ultrafilters of } B \,\}, \\ h &\longmapsto h^{-1}[\{\, 1 \,\}], \\ \left(\chi_U \colon x \mapsto \begin{cases} 1, & x \in U, \\ 0, & x \notin U, \end{cases}\right) &\longleftarrow\!\shortmid U. \end{aligned}$$

Proof sketch. If $h \colon B \to 2$ is a homomorphism, set $U \coloneqq h^{-1}[\{\, 1 \,\}]$. Then $1 \in U$, U is upward closed and closed under meets, so it is a filter; it is proper since $h(0) = 0$. Moreover, for each $x \in B$, $h(\neg x) = \neg h(x)$, and hence exactly one of x and $\neg x$ lands in 1. Thus, U satisfies the conditions in Lemma 1.3.6 and so is an ultrafilter.

Conversely, if U is an ultrafilter, define $h_U \colon B \to 2$ by $h_U(x) = 1$ if and only if $x \in U$. The ultrafilter axioms ensure that h_U preserves $\wedge$, $\neg$, and 1, hence h_U is a homomorphism. ■

Therefore, ultrafilters of B are encodings of homomorphisms from B to 2. Thus, we will use the set of ultrafilters of B as the set X such that B embeds into $\mathcal{P}(X)$.

Definition 1.3.8 (Principal filter)**.** The ***principal filter generated by an element*** a of a Boolean algebra B is

$$\uparrow a \;\coloneqq\; \{\, x \in B \mid a \leq x \,\}.$$

Therefore, a filter (and, in particular, an ultrafilter) is called ***principal*** if it has a minimum.

Remark 1.3.9. If B is finite, then every filter is principal.

We will soon see that $\uparrow a$ is an ultrafilter if and only if a satisfies the following simple condition.

Definition 1.3.10 (Atom)**.** An ***atom*** of a Boolean algebra B is a minimal element of $B \setminus \{\, 0 \,\}$.

Proposition 1.3.11 (Atoms and ultrafilters)**.**

1. *For every Boolean algebra B we have a bijection*

$$\begin{aligned} \{\, \textit{atoms of } B \,\} &\longleftrightarrow \{\, \textit{principal ultrafilters of } B \,\} \\ a &\longmapsto \uparrow a \\ \min U &\longleftarrow\!\shortmid U. \end{aligned} \tag{*}$$

2. *If B is <u>finite</u>, every ultrafilter is principal and so $*$ gives a bijection between atoms and ultrafilters.*

Proof. We only need to prove that the two functions are well defined; that they are mutually inverse will then be immediate.

To prove that the left-to-right function is well defined, let us assume that a is an atom, and let us prove that $U \coloneqq \uparrow a$ is an ultrafilter. It is a filter (easy). $0 \notin U$ because $a > 0$. Suppose $x \vee y \in \uparrow a$, i.e., $x \vee y \geq a$. Then, by distributivity,

$$a = a \wedge (x \vee y) = (a \wedge x) \vee (a \wedge y).$$

Therefore, since the only elements below a are 0 and a, and the join of $(a \wedge x)$ and $(a \wedge y)$ is a, at least one of the two is a (otherwise they would both be 0 and so their join would be 0). If $a \wedge x = a$ then $x \in \uparrow a$, and if $a \wedge y = a$ then $y \in \uparrow a$. Thus, U is an ultrafilter.

To prove that the right-to-left function is well defined, let us assume that $\uparrow a$ is a principal ultrafilter, and let us prove that $\min U$ is an atom. We have $a \neq 0$ (otherwise 0 would belong to U). If $0 < x < a$, then $x \notin \uparrow a$, and hence $\neg x \in \uparrow a$ by Lemma 1.3.6. Thus $a \leq \neg x$, and since $x \leq a$ we get $x \leq \neg x$, hence $x = x \wedge \neg x = 0$, a contradiction. Therefore, there is no x with $0 < x < a$, so a is an atom. ■

Example 1.3.12. The trivial Boolean algebra of cardinality 1 has no ultrafilters (and no homomorphisms to 2). The Boolean algebra $2 = \{ 0, 1 \}$ has exactly one ultrafilter, namely $\{ 1 \}$. The Boolean algebra of cardinality 4 has exactly two ultrafilters, corresponding to its two atoms. The Boolean algebra of cardinality 8 has exactly three ultrafilters, corresponding to its three atoms.

In an infinite Boolean algebra, an ultrafilter may fail to be principal, i.e., to have a minimum. The following gives an example.

Example 1.3.13 (Ultrafilters of $\mathrm{FC}(\mathbb{N})$)**.** For each $n \in \mathbb{N}$, the set

$$U_n \;\coloneqq\; \{ A \in \mathrm{FC}(\mathbb{N}) \mid n \in A \}$$

is a (principal) ultrafilter on the finite-cofinite algebra $\mathrm{FC}(\mathbb{N})$ on $\mathbb{N}$.

In addition, the family of cofinite sets

$$U_\infty \;\coloneqq\; \{ A \in \mathrm{FC}(\mathbb{N}) \mid A \text{ is cofinite} \}$$

is an ultrafilter on $\mathrm{FC}(\mathbb{N})$; it is not principal, since it has no minimum. As a simple exercise, you can prove that it has no other ultrafilters, i.e., that

$$\mathrm{Ult}(\mathrm{FC}(\mathbb{N})) = \{ U_n \mid n \in \mathbb{N} \} \cup \{ U_\infty \}.$$

A proof is in the footnote.[6]

Remark 1.3.14 (Ultrafilters on a power set)**.** What are the ultrafilters of $\mathcal{P}(X)$, for X a set? First of all, for every $x \in X$, we have an ultrafilter

$$U_x \coloneqq \{ A \in \mathcal{P}(X) \mid x \in A \}.$$

These are precisely the *principal* ultrafilters.

- If X is finite, these are the only ultrafilters (see Proposition 1.3.11).
- For X infinite, does $\mathcal{P}(X)$ have nonprincipal ultrafilters? This is a nontrivial question; we will see that, if we assume the axiom of choice, we can prove the existence of nonprincipal ultrafilters; this will follow from the Boolean Prime Ideal Theorem (Theorem 1.3.17) proved below. Do not hope to find an explicit description of any of them, though.

1.3.2 Stone's Representation Theorem

Let us now take $X = \mathrm{Ult}(B)$ and define the embedding $B \hookrightarrow \mathcal{P}(X)$ required by Stone's Representation Theorem. Intuitively, this sends an element b to the set of "models" (ultrafilters) where b holds.

[6] Let U be an ultrafilter on $\mathrm{FC}(\mathbb{N})$. If U contains a finite set A, write A as a finite union of singletons: $A = \{ n_1 \} \cup \cdots \cup \{ n_k \}$. Since U is an ultrafilter, the primeness condition in Definition 1.3.5 implies that some singleton $\{ n_i \}$ belongs to U. Then U must coincide with U_{n_i}: if $B \in \mathrm{FC}(\mathbb{N})$ contains n_i, then $\{ n_i \} \subseteq B$, hence $B \in U$ by upward closure; if $n_i \notin B$, then $n_i \in \mathbb{N} \setminus B$ and $\mathbb{N} \setminus B \in U$, hence $B \notin U$. If instead U contains no finite set, then for every finite $F \subseteq \mathbb{N}$ we have $F \notin U$, hence $\mathbb{N} \setminus F \in U$ by Lemma 1.3.6. Thus U contains every cofinite set, i.e. $U_\infty \subseteq U$. Since U_∞ is itself an ultrafilter, maximality forces $U = U_\infty$.

Definition 1.3.15 (Stone map)**.** For a Boolean algebra B, we define the ***Stone map***

$$\begin{aligned} \eta_B \colon B &\longrightarrow \mathcal{P}(\mathrm{Ult}(B)) \\ b &\longmapsto \{\, U \in \mathrm{Ult}(B) \mid b \in U \,\}. \end{aligned}$$

Logical intuition: the set $\eta_B(b)$ can be thought of as the set of "models" in which the "formula" b holds. To prove Stone's Representation Theorem it suffices to show that:

1. η_B is a Boolean homomorphism (easy),
2. η_B is injective (hard).

Lemma 1.3.16 (The Stone map is a homomorphism)**.** *The Stone map η_B is a Boolean homomorphism, i.e. for all $a, b \in B$,*

$$\eta_B(1) = \mathrm{Ult}(B), \qquad \eta_B(a \wedge b) = \eta_B(a) \cap \eta_B(b), \qquad \eta_B(\neg a) = \mathrm{Ult}(B) \setminus \eta_B(a),$$

(and hence also $\eta_B(a \vee b) = \eta_B(a) \cup \eta_B(b)$ and $\eta_B(0) = \varnothing$).

Proof. These are all straightforward computations. Details are in the footnote.[a] ■

[a] The equality $\eta_B(1) = \mathrm{Ult}(B)$ is immediate: every ultrafilter contains 1.
For the binary meet, let $U \in \mathrm{Ult}(B)$. Then

$$U \in \eta_B(a \wedge b) \iff a \wedge b \in U \iff (a \in U \text{ and } b \in U) \iff U \in \eta_B(a) \cap \eta_B(b),$$

where the middle equivalence uses upward closure and closure under binary meets of U.
For complements, using Lemma 1.3.6 we have

$$U \in \eta_B(\neg a) \iff \neg a \in U \iff a \notin U \iff U \notin \eta_B(a).$$

To conclude the proof of Stone's Representation Theorem (Theorem 1.3.20), it remains to show that η_B is injective. This is a separation statement: given two distinct elements of B, we must find an ultrafilter containing one but not the other. The required separation will follow from the Boolean Prime Ideal Theorem, proved next.

Let us recall Zorn's Lemma:

> Let $(P, \leq)$ be a partially ordered set. If
>
> 1. $P \neq \varnothing$,
> 2. and every nonempty chain[7] in P has an upper bound in P,
>
> then P has a maximal element.

Recall that Zorn's lemma is not provable in ZF, and is equivalent to the Axiom of Choice over ZF. We will use it in the following, which in turn we will use to conclude the proof of Stone's Representation Theorem.

Can every partial consistent choice of truth values be extended to a total one? The following can be thought of as a positive answer.

Theorem 1.3.17 (Boolean Prime Ideal Theorem)**.** *Let B be a Boolean algebra, let F be a filter, and let I be an ideal such that $F \cap I = \varnothing$. Then there is an ultrafilter U on B such that $F \subseteq U$ and $U \cap I = \varnothing$.*

[7] A chain is a totally ordered poset. Thus, a *chain in P* is a subset S of P such that, for all $x, y \in S$, either $x \leq y$ or $y \leq x$.

Proof. Consider the poset

$$\mathcal{P} \;:=\; \{\,(G,J) \mid G \text{ is a filter},\ J \text{ is an ideal},\ F \subseteq G,\ I \subseteq J,\ G \cap J = \varnothing\,\},$$

ordered by componentwise inclusion: $(G,J) \leq (G',J')$ iff $G \subseteq G'$ and $J \subseteq J'$.

Nonemptiness. This poset is nonempty since $(F,I) \in \mathcal{P}$.

Nonempty chains have upper bounds. Let $\mathcal{C} \subseteq \mathcal{P}$ be a nonempty chain and set

$$G^* := \bigcup_{(G,J)\in\mathcal{C}} G, \qquad J^* := \bigcup_{(G,J)\in\mathcal{C}} J.$$

It is straightforward to prove that $(G^*, J*)$ belongs to $\mathcal{P}$: one checks that G^* is a filter, J^* is an ideal, $F \subseteq G^*$, $I \subseteq J^*$ and $G^* \cap J^* = \varnothing$. See the footnote for details. Thus $(G^*, J^*) \in \mathcal{P}$ is an upper bound of $\mathcal{C}$.

By Zorn's lemma, $\mathcal{P}$ has a maximal element (U,K). At this point, the key step is to show that the maximal pair (U,K) "splits" B: once we know that $U \cup K = B$, the disjointness $U \cap K = \varnothing$ forces $K = B \setminus U$. Since K is an ideal, this will imply that $B \setminus U$ is an ideal, and hence U is an ultrafilter.

We claim that $U \cup K = B$. Suppose not, and pick $x \in B \setminus (U \cup K)$.

Let U' be the filter generated by $U \cup \{\,x\,\}$. Concretely,

$$U' = \{\,y \in B \mid \exists u \in U,\ (u \wedge x) \leq y\,\}.$$

If $U' \cap K = \varnothing$, then $(U',K) \in \mathcal{P}$ strictly extends (U,K), contradicting maximality. Hence, there are $u \in U$ and $k \in K$ such that $u \wedge x \leq k$, and thus $u \wedge x \in K$.

Similarly, let K' be the ideal generated by $K \cup \{\,x\,\}$. Concretely,

$$K' = \{\,y \in B \mid \exists k \in K,\ y \leq (k \vee x)\,\}.$$

If $U \cap K' = \varnothing$, then $(U,K') \in \mathcal{P}$ strictly extends (U,K), again contradicting maximality. Hence, there are $u' \in U$ and $k' \in K$ such that $u' \leq k' \vee x$.

Set $w := u \wedge u' \in U$. Then $w \wedge x \leq u \wedge x \in K$, hence $w \wedge x \in K$ (since K is downward closed), and also $w \leq u' \leq k' \vee x$. Using distributivity,

$$w = w \wedge (k' \vee x) = (w \wedge k') \vee (w \wedge x).$$

Now $w \wedge k' \leq k'$, hence $w \wedge k' \in K$, and we already know $w \wedge x \in K$. Since K is closed under finite joins, we conclude $w \in K$. This contradicts $w \in U$ and $U \cap K = \varnothing$. Therefore $U \cup K = B$. This proves that U is an ultrafilter. It is clear that $F \subseteq U$ and $U \cap I = \varnothing$. ■

Corollary 1.3.18 (Ultrafilters separate elements)**.** *If $a \not\leq b$ in B, then there is an ultrafilter $U \in \mathrm{Ult}(B)$ such that $a \in U$ and $b \notin U$.*

Proof. Let $F := \uparrow a$ be the principal filter generated by a, and let

$$I := \downarrow b := \{\,x \in B \mid x \leq b\,\}$$

be the principal ideal generated by b. We notice that $F \cap I = \varnothing$, since otherwise there would be $x \in \uparrow a \cap \downarrow b$, which would imply $a \leq x \leq b$, which would contradict $a \not\leq b$. By Theorem 1.3.17 there is an ultrafilter U such that $F \subseteq U$ and $U \cap I = \varnothing$. In particular, $a \in U$ and $b \notin U$. ■

As a consequence:

Lemma 1.3.19 (The Stone map is injective)**.** *For every Boolean algebra B, the Stone map $\eta_B \colon B \to \mathcal{P}(\mathrm{Ult}(B))$ defined in Definition 1.3.15 is injective.*

Proof. Let $a \neq b$ in B. Then either $a \not\leq b$ or $b \not\leq a$. Assume $a \not\leq b$. (The proof in the other case is perfectly symmetrical.) By Corollary 1.3.18, there is $U \in \mathrm{Ult}(B)$ with $a \in U$ and $b \notin U$. Thus $U \in \eta_B(a)$ but $U \notin \eta_B(b)$, so $\eta_B(a) \neq \eta_B(b)$. ∎

We are now ready to prove Stone's Representation Theorem for Boolean algebras [Stone, 1936], which is a positive answer to the question: can every abstract Boolean algebra be made concrete by representing its elements as actual subsets of some set?

Theorem 1.3.20 (Stone's Representation Theorem for Boolean algebras)**.** *For every Boolean algebra B there is a set X such that B is isomorphic to a Boolean subalgebra of the power set $\mathcal{P}(X)$.*

Proof. This follows from the fact that the Stone map $\eta_B \colon B \to \mathcal{P}(\mathrm{Ult}(B))$ is a Boolean homomorphism (Lemma 1.3.16) and is injective (Lemma 1.3.19). ∎

Remark 1.3.21 (Choice)**.** To prove Stone's Representation Theorem we used the axiom of choice (since we used Zorn's lemma to prove the Boolean Prime Ideal Theorem). In fact, Stone's Representation Theorem is not provable in ZF alone. Over ZF, Stone's Representation Theorem for Boolean algebras is equivalent to the Boolean Prime Ideal Theorem; see, for instance, [Jech, 1973, Sec. 2.6]. The Boolean Prime Ideal Theorem is weaker than the Axiom of Choice, in the sense that it follows from Choice (via Zorn's lemma), but it does not imply Choice [Halpern and Lévy, 1971].

1.4 Boolean algebras are precisely the Lindenbaum–Tarski algebras

In Section 1.2.4 we saw that every propositional theory $\mathcal{T}$ gives rise to a Boolean algebra $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$, where $\equiv_{\mathcal{T}}$ denotes the semantic equivalence between formulas modulo $\mathcal{T}$: this Boolean algebra is called the Lindenbaum–Tarski algebra of $\mathcal{T}$.

Now we prove the converse statement announced earlier: *every Boolean algebra arises in this way.*

Warm-up examples

Example 1.4.1 (2 as a Lindenbaum–Tarski algebra)**.** Let $\mathcal{L} = \varnothing$ and $\mathcal{T} = \varnothing$. Then $\mathrm{Form}(\mathcal{L})$ only contains the constants $0, 1$ and the formulas built from them. Up to $\equiv_{\mathcal{T}}$ there are exactly two equivalence classes, so $\mathrm{Form}(\varnothing)/{\equiv_{\varnothing}} \cong 2$.

Example 1.4.2 (The 4-elements Boolean algebra as a Lindenbaum–Tarski algebra)**.** Let $\mathcal{L} = \{\, p \,\}$ and $\mathcal{T} = \varnothing$. Then $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ is a Boolean algebra with 4 elements (it is isomorphic to $\mathcal{P}(\{x, y\})$).

Example 1.4.3 (The trivial Boolean algebra as a Lindenbaum–Tarski algebra)**.** If a theory $\mathcal{T}$ has no models (i.e. $\mathrm{Mod}(\mathcal{T}) = \varnothing$), then by definition every two formulas are $\equiv_{\mathcal{T}}$-equivalent. Hence,Hence, $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ is the trivial Boolean algebra. For instance, one can take any $\mathcal{L}$ and $\mathcal{T} = \{\, 0 \,\}$.

Every Boolean algebra is a Lindenbaum–Tarski algebra

We now show that *every* Boolean algebra can be presented as $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ for a suitable language $\mathcal{L}$ and theory $\mathcal{T}$, i.e., that it is the algebra of propositions of a suitable language and theory.

Theorem 1.4.4 (Every Boolean algebra is a Lindenbaum–Tarski algebra)**.** *For every Boolean algebra B*

there is a propositional language $\mathcal{L}$ and a propositional theory $\mathcal{T}$ such that

$$B \;\cong\; \mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}.$$

Proof. Let B be a Boolean algebra. We build a language whose propositional variables are *the elements of B themselves*:

$$\mathcal{L}_B \;\coloneqq\; B.$$

To avoid confusion, we will use $\wedge_B, \vee_B, \neg_B, 0_B, 1_B$ for the operations of the Boolean algebra B, while $\wedge, \vee, \neg, 0, 1$ are the connectives/constants in $\mathrm{Form}(\mathcal{L}_B)$.

Define a theory $\mathcal{T}_B$ that forces the propositional variables to behave like B:

$$\mathcal{T}_B \;\coloneqq\; \Big\{\, (\neg b) \leftrightarrow (\neg_B b) \;\Big|\; b \in B \,\Big\} \;\cup\; \Big\{\, (b \wedge c) \leftrightarrow (b \wedge_B c) \;\Big|\; b, c \in B \,\Big\} \;\cup\; \{\, 1 \leftrightarrow 1_B \,\}.$$

(Here $(\neg_B b)$, $(b \wedge_B c)$ and 1_B are elements of B, and hence propositional variables in the language $\mathcal{L}_B$. Moreover, the axioms for $\vee$ and 0 are not necessary since $\vee$ and 0 are definable from $\wedge, \neg, 1$.)

Now consider the assignment $e\colon \mathcal{L}_B \to B$ given by the identity map $e(b) = b$. As for truth assignments into 2, the map e extends uniquely (by recursion on formulas)[a] to an evaluation map

$$\bar{e}\colon \mathrm{Form}(\mathcal{L}_B) \to B,$$

defined by interpreting $\vee, \wedge, \neg, 0, 1$ as $\vee_B, \wedge_B, \neg_B, 0_B, 1_B$. In particular, $\bar{e}$ is a $\{\vee, \wedge, \neg, 0, 1\}$-homomorphism, and it is surjective because every $b \in B$ is the value of the variable b.

We claim that for all formulas $\varphi, \psi \in \mathrm{Form}(\mathcal{L}_B)$ we have

$$\varphi \equiv_{\mathcal{T}_B} \psi \qquad \Longleftrightarrow \qquad \bar{e}(\varphi) = \bar{e}(\psi) \;\text{ in } B.$$

Granting this for a moment, the isomorphism $B \cong \mathrm{Form}(\mathcal{L}_B)/{\equiv_{\mathcal{T}_B}}$ follows immediately: define $\pi\colon \mathrm{Form}(\mathcal{L}_B) \twoheadrightarrow \mathrm{Form}(\mathcal{L}_B)/{\equiv_{\mathcal{T}_B}}$ and set

$$\Phi\colon B \to \mathrm{Form}(\mathcal{L}_B)/{\equiv_{\mathcal{T}_B}}, \qquad \Phi(b) \coloneqq [\, b \,].$$

The claim says precisely that Φ is injective (two generators $[\, b \,], [\, c \,]$ coincide iff $b = c$ in B), and surjectivity is clear because every class is represented by some formula and $\bar{e}$ hits all elements of B. Moreover, Φ is a Boolean homomorphism by construction. For instance, $\Phi(b \wedge_B c) = [\, b \wedge_B c \,] = [\, b \wedge c \,] = [\, b \,] \wedge [\, c \,] = \Phi(b) \wedge \Phi(c)$ by definition of $\mathcal{T}_B$, and similarly for $\neg$ and 1. Hence, Φ is an isomorphism.

It remains to prove the claim. First, observe the key fact (which is a direct unpacking of the axioms of $\mathcal{T}_B$):

> *Models of $\mathcal{T}_B$ are the same thing as Boolean homomorphisms $h\colon B \to 2$.*

Indeed, if $h\colon B \to 2$ is a Boolean homomorphism, then the valuation $v_h\colon \mathcal{L}_B \to 2$ given by $v_h(b) = h(b)$ satisfies the axioms of $\mathcal{T}_B$, so $v_h \in \mathrm{Mod}(\mathcal{T}_B)$. Conversely, if $v \in \mathrm{Mod}(\mathcal{T}_B)$, then the map $h_v\colon B \to 2$ defined by $h_v(b) \coloneqq v(b)$ is a Boolean homomorphism, because the axioms in $\mathcal{T}_B$ exactly say that v respects $\neg_B$, $\wedge_B$, and 1_B.

Now suppose $\bar{e}(\varphi) = \bar{e}(\psi)$. Let $v \in \mathrm{Mod}(\mathcal{T}_B)$ and let $h_v\colon B \to 2$ be the corresponding homomorphism. By construction of evaluation-by-recursion, the interpretation of formulas agrees:

$$\bar{v} \;=\; h_v \circ \bar{e}.$$

Therefore $\bar{v}(\varphi) = h_v(\bar{e}(\varphi)) = h_v(\bar{e}(\psi)) = \bar{v}(\psi)$. Since this holds for all $v \in \mathrm{Mod}(\mathcal{T}_B)$, we get $\varphi \equiv_{\mathcal{T}_B} \psi$.

Conversely, suppose $\bar{e}(\varphi) \neq \bar{e}(\psi)$ in B. By the fact that ultrafilters separate distinct elements of B (i.e., that $\eta_B \colon B \to \mathcal{P}(\mathrm{Ult}(B))$ is injective, Lemma 1.3.19), there is a Boolean homomorphism $h \colon B \to 2$ such that

$$h(\bar{e}(\varphi)) \;\neq\; h(\bar{e}(\psi)).$$

Let $v_h \in \mathrm{Mod}(\mathcal{T}_B)$ be the corresponding model. Then again $\bar{v}_h = h \circ \bar{e}$, hence $\bar{v}_h(\varphi) \neq \bar{v}_h(\psi)$, so $\varphi \not\equiv_{\mathcal{T}_B} \psi$. This proves the claim and completes the proof. ■

[a] Universal-algebra viewpoint: for the algebraic signature $\Sigma = \{0, 1, \neg, \wedge, \vee\}$, the forgetful functor $\mathsf{Alg}(\Sigma) \to \mathsf{Set}$ has a left adjoint sending a set $\mathcal{L}$ to the Σ-algebra of terms on $\mathcal{L}$. Here, $\bar{e}$ is exactly the unique Σ-homomorphism extending e, whose existence is guaranteed by the universal property of the unit of the adjunction.

1.5 Bijection between Boolean algebras and Stone spaces

Stone's Representation Theorem says that every Boolean algebra embeds into a power set. However, such a representation is typically *not unique*. There are two properties—*separation* and *compactness*—that single out the canonical representation

$$\eta_B \colon B \hookrightarrow \mathcal{P}(\mathrm{Ult}(B)).$$

Example 1.5.1 (Separated vs non-separated)**.** For instance, the two-element Boolean algebra 2 has the following two representations:

$$\begin{aligned} \iota_1 \colon 2 &\hookrightarrow \mathcal{P}(\{*\}) \\ 0 &\longmapsto \varnothing \\ 1 &\longmapsto \{*\} \end{aligned} \qquad\qquad \begin{aligned} \iota_2 \colon 2 &\hookrightarrow \mathcal{P}(\{x, y\}) \\ 0 &\longmapsto \varnothing \\ 1 &\longmapsto \{x, y\}. \end{aligned}$$

The representation ι_2 is not "*separated*": the only sets in its image are $\varnothing$ and $\{x, y\}$, so the image cannot distinguish x from y.

Example 1.5.2 (Non-compact vs compact)**.** Let $\mathrm{FC}(\mathbb{N})$ be the Boolean algebra of finite and cofinite subsets of $\mathbb{N}$. It has the following two representations:

$$\begin{aligned} \jmath_1 \colon \mathrm{FC}(\mathbb{N}) &\hookrightarrow \mathcal{P}(\mathbb{N}) \\ A &\longmapsto A \end{aligned} \qquad\qquad \begin{aligned} \jmath_2 \colon \mathrm{FC}(\mathbb{N}) &\hookrightarrow \mathcal{P}(\mathbb{N} \cup \{\infty\}) \\ A &\longmapsto \begin{cases} A, & \text{if } A \text{ is finite,} \\ A \cup \{\infty\}, & \text{if } A \text{ is cofinite.} \end{cases} \end{aligned}$$

The representation $\jmath_1$ is not "*compact*": the family of singletons $\{\{n\} \mid n \in \mathbb{N}\}$ consists of elements of $\mathrm{FC}(\mathbb{N})$ and covers $\mathbb{N}$, but it has no finite subcover.

On the other hand, $\jmath_2$ is *compact*. Indeed, let $\{\jmath_2(A_i)\}_{i \in I}$ be a cover of $\mathbb{N} \cup \{\infty\}$. Since ∞ is covered, there is $i_0 \in I$ such that $\infty \in \jmath_2(A_{i_0})$. By definition of $\jmath_2$, this means that A_{i_0} is cofinite in $\mathbb{N}$. Then $\jmath_2(A_{i_0})$ already covers all but finitely many natural numbers; the remaining finitely many points of $\mathbb{N}$ can be covered by finitely many singleton sets $\jmath_2(\{n\}) = \{n\}$. Hence, the original cover admits a finite subcover.

The previous examples show that representations into power sets can fail to be *separated* or *compact*. Spoiler: for every Boolean algebra B, the canonical representation

$$\eta_B \colon B \hookrightarrow \mathcal{P}(\mathrm{Ult}(B))$$

is (up to renaming the points of the underlying set) the unique one that is *both* separated and compact.

Now, both separation and compactness are strongly "topology-like" properties. So rather than treating them as ad hoc conditions on set-theoretic representations, we move to the topological setting: we equip $\mathrm{Ult}(B)$ with a natural topology generated by the basic subsets $\eta_B(b)$. With this topology, $\mathrm{Ult}(B)$ becomes a *Stone space*, i.e. a space that is totally separated and compact (definitions to follow).

Then, we will get a bijection between isomorphism classes of Boolean algebras and isomorphism classes of Stone spaces.

1.5.1 Stone spaces and the Boolean algebra of clopens

Let X be a topological space. A subset $C \subseteq X$ is called ***clopen*** if it is both closed and open in X. We write

$$\mathrm{Clop}(X)$$

for the set of clopen subsets of X. Notice that $\mathrm{Clop}(X)$ is a Boolean subalgebra of $\mathcal{P}(X)$.

Definition 1.5.3 (Stone space)**.** A ***Stone space***[a] is a topological space that is

1. ***totally separated***: for all $x, y \in X$ with $x \neq y$ there is a clopen subset C of X such that $x \in C$ and $y \notin C$;
2. ***compact***: every open cover of X has a finite subcover.

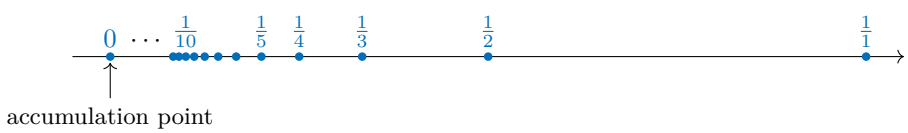
[a] Two further equivalent characterizations of Stone spaces are: (i) compact Hausdorff zero-dimensional spaces, and (ii) compact Hausdorff totally disconnected spaces.

Notice that totally separated implies Hausdorff.

Example 1.5.4 (Examples of Stone spaces)**.**

1. (Finite discrete spaces) Every *finite discrete space* is a Stone space. In fact, these are the only finite Stone spaces.
2. (One-point compactification of $\mathbb{N}$) The simplest example of an infinite Stone space is called the *one-point* (or *Alexandroff*) *compactification* of $\mathbb{N}$. It is denoted $\alpha\mathbb{N}$, and is obtained by adding to the discrete space $\mathbb{N}$ an accumulation point ∞. More precisely, equip $\mathbb{N} \cup \{\infty\}$ with the topology whose open sets are:
   - every subset of $\mathbb{N}$, and
   - every cofinite subset U such that $\infty \in U$.

   For a visual rendering, this is homeomorphic to the following subspace of $\mathbb{R}$ via the map $n \mapsto \frac{1}{n}$ and $\infty \mapsto 0$:
   $$\{0\} \cup \left\{ \frac{1}{n} \mid n \in \mathbb{N} \setminus \{0\} \right\}.$$



Example 1.5.5 (*Non*-examples of Stone spaces)**.**

1. Any infinite discrete space is *not* a Stone space, because it is not compact.
2. $[0, 1]$ is *not* a Stone space, because it is not totally separated.

1.5.2 The Stone topology on the set of ultrafilters

Definition 1.5.6 (Stone topology on the set of ultrafilters)**.** Let B be a Boolean algebra. The ***Stone topology*** on $\mathrm{Ult}(B)$ is the topology *generated* by the subsets

$$\eta_B(b) \qquad (b \in B).$$

That is, the open sets are the arbitrary unions of elements of the form[a]

$$\eta_B(b) \qquad (b \in B).$$

[a] In general, the topology generated by a set $\mathcal{A}$ of subsets of a set X is the set of arbitrary unions of finite intersections of elements in $\mathcal{A}$. Here, there is no need to take finite intersections because the family $\{\, \eta_B(b) \mid b \in B \,\}$ is closed under finite intersections.

Thus, the family $\{\, \eta_B(b) \mid b \in B \,\}$ forms a basis of the Stone topology on $\mathrm{Ult}(B)$, and hence the elements of the form $\eta_B(b)$ are called *basic open* sets of $\mathrm{Ult}(B)$.

Theorem 1.5.7 (The space of ultrafilters is a Stone space)**.** *For every Boolean algebra B, the space* $\mathrm{Ult}(B)$ *of ultrafilters equipped with the Stone topology is a Stone space.*

Proof. Totally separated. Let $U, V \in \mathrm{Ult}(B)$ with $U \neq V$. Then, there is b which belongs to exactly one of the two, say, for example, U (the other case is perfectly symmetrical). Then $U \in \eta_B(b)$ and $V \notin \eta_B(b)$. Moreover, $\eta_B(b)$ is clopen, since its complement is $\eta_B(\neg b)$, which is open. Hence, $\eta_B(b)$ is a clopen separating U from V. Thus, $\mathrm{Ult}(B)$ is totally separated.

Compact. To check the compactness of a space, it is enough to check that any cover by opens in a given basis has a finite subcover. Suppose that $\mathrm{Ult}(B)$ is covered by basic opens:

$$\mathrm{Ult}(B) \;=\; \bigcup_{j \in J} \eta_B(b_j). \tag{1.2}$$

Consider the ideal $I \subseteq B$ generated by the set $\{\, b_j \mid j \in J \,\}$, i.e.

$$I = {\downarrow}\{\, b_{j_1} \vee \cdots \vee b_{j_n} \mid n \in \mathbb{N}, j_1, \ldots, j_n \in J \,\}.$$

We claim that $1 \in I$. Indeed, if, by way of contradiction, we had $1 \notin I$, then the filter $\{1\}$ would be disjoint from I; then, by the Boolean Prime Ideal Theorem (Theorem 1.3.17), there would be an ultrafilter U extending $\{1\}$ and disjoint from I. But by hypothesis ((1.2)), we deduce that there is $j \in J$ such that $U \in \eta_B(b_j)$, i.e., $b_j \in U$. This contradicts the facts that $b_j \in I$ and that I and U are disjoint. This proves the claim $1 \in I$.

Since $1 \in I$, by definition of I there are $j_1, \ldots, j_n \in J$ such that

$$1 \;\leq\; b_{j_1} \vee \cdots \vee b_{j_n},$$

and hence

$$1 \;=\; b_{j_1} \vee \cdots \vee b_{j_n}.$$

Applying η_B on both sides, and using the fact that it η_B a Boolean homomorphism (Lemma 1.3.16), we get

$$\mathrm{Ult}(B) = \eta_B(1) = \eta_B(b_{j_1} \vee \cdots \vee b_{j_n}) = \eta_B(b_{j_1}) \cup \cdots \cup \eta_B(b_{j_n}),$$

and so we have found a finite subcover. ∎

In the proof of Theorem 1.5.7 we used the (immediate) fact that the subsets of $\mathrm{Ult}(B)$ of the form $\eta_B(b)$ are clopen; in fact, the sets of the form $\eta_B(b)$ are all and the only clopens, as shown in the following.

Lemma 1.5.8 (In the space of ultrafilters: clopen = basic open)**.** *Let B be a Boolean algebra. The clopens of* $\mathrm{Ult}(B)$ *are precisely the subsets of the form $\eta_B(b)$, for some $b \in B$.*

Proof. ($\Leftarrow$). For every $b \in B$, $\eta_B(b)$ is clopen because it is open and its complement is $\eta_B(\neg b)$ (since η_B is a Boolean homomorphism by Lemma 1.3.16), which is open, too.

($\Rightarrow$). Let C be a clopen of $\mathrm{Ult}(B)$. Since C is open, we can write

$$C = \bigcup_{i \in I} \eta_B(b_i)$$

for some family $(b_i)_{i \in I}$ in B, and automatically each $\eta_B(b_i) \subseteq C$. Because C is closed in the space $\mathrm{Ult}(B)$, which is compact by Theorem 1.5.7, the subspace C is compact. Thus the open cover $\{\, \eta_B(b_i) \mid i \in I \,\}$ of C has a finite subcover: there is a finite $I_0 \subseteq I$ such that

$$C = \bigcup_{i \in I_0} \eta_B(b_i).$$

Since η_B is a homomorphism,

$$C = \eta_B\left(\bigvee_{i \in I_0} b_i \right),$$

so C lies in the image of η_B. ∎

1.5.3 Bijection between Boolean algebras and Stone spaces

Theorem 1.5.9 ($B \cong \mathrm{Clop}(\mathrm{Ult}(B))$)**.** *For every Boolean algebra B, the map*

$$\begin{aligned} \eta_B \colon B &\longrightarrow \mathrm{Clop}(\mathrm{Ult}(B)), \\ b &\longmapsto \{\, U \in \mathrm{Ult}(B) \mid b \in U \,\} \end{aligned}$$

is an isomorphism of Boolean algebras.

Proof. By Lemma 1.5.8, for every b the set $\eta_B = \{\, U \in \mathrm{Ult}(B) \mid b \in U \,\}$ is indeed clopen, and the map $\eta_B \colon B \longrightarrow \mathrm{Clop}(\mathrm{Ult}(B))$ is surjective. By Lemmas 1.3.16 and 1.3.19, $\eta_B \colon B \longrightarrow \mathrm{Clop}(\mathrm{Ult}(B))$ is an injective homomorphism. ∎

It follows immediately:

Corollary 1.5.10 (Finite Boolean algebras are power sets)**.**

1. *Every <u>finite</u> Boolean algebra is isomorphic to the power set of some finite set.*
2. *Every <u>finite</u> Boolean algebra has cardinality 2^n for some $n \in \mathbb{N}$.*
3. *Any two <u>finite</u> Boolean algebras with the same cardinality are isomorphic.*

Proof. (1). Let B be finite. Then $\mathrm{Ult}(B)$ is finite as well (indeed $\mathrm{Ult}(B) \subseteq \mathcal{P}(B)$). Since $\mathrm{Ult}(B)$ is a finite Stone space, it is discrete[a], so every subset is clopen and hence

$$\mathrm{Clop}(\mathrm{Ult}(B)) = \mathcal{P}(\mathrm{Ult}(B)).$$

By the isomorphism $B \cong \mathrm{Clop}(\mathrm{Ult}(B))$ we conclude that $B \cong \mathcal{P}(X)$ for the finite set $X \coloneqq \mathrm{Ult}(B)$.

(2). This follows immediately from (1).

(3). Let B and B' be finite Boolean algebras and assume $|B| = |B'|$. By (1), B and B' are isomorphic

to $\mathcal{P}(X)$ and $\mathcal{P}(X')$, respectively, for some finite sets X and X'. We have

$$2^{|X|} = |B| = |B'| = 2^{|X'|},$$

and so $|X| = |X'|$. It is immediate that two sets with the same cardinality have isomorphic power set Boolean algebras; hence, $B \cong \mathcal{P}(X) \cong \mathcal{P}(X') \cong B'$. ■

[a]It is easily seen that every T_1 finite space (and so, in particular, every Stone space) is discrete.

Theorem 1.5.11 ($X \cong \mathrm{Ult}(\mathrm{Clop}(X))$)**.** *For every Stone space X, the map*

$$\begin{aligned} \varepsilon_X \colon X &\longrightarrow \mathrm{Ult}(\mathrm{Clop}(X)) \\ x &\longmapsto \{\, C \in \mathrm{Clop}(X) \mid x \in C \,\} \end{aligned}$$

is a homeomorphism (where $\mathrm{Ult}(\mathrm{Clop}(X))$ *carries the Stone topology).*

Proof. First of all, the map is well-defined because, for every $x \in X$, the set $\varepsilon_X(x) = \{\, C \in \mathrm{Clop}(X) \mid x \in C \,\}$ is easily seen[a] to be an ultrafilter of $\mathrm{Clop}(X)$.

Since $\varepsilon_X \colon X \to \mathrm{Ult}(\mathrm{Clop}(X))$ is a function between compact Hausdorff spaces, to prove that it is a homeomorphism it is enough[b] to prove that it is

1. continuous (this is a straightforward computation),
2. injective (this will follow from <u>total separation</u>),
3. surjective (this will follow from <u>compactness</u>).

Continuity. This is a straightforward computation.[c]

Injectivity. If $x \neq y$, since X is <u>totally separated</u> there is a clopen $C \subseteq X$ such that $x \in C$ and $y \notin C$. Then $C \in \varepsilon_X(x)$ but $C \notin \varepsilon_X(y)$, and hence $\varepsilon_X(x) \neq \varepsilon_X(y)$.

Surjectivity. Let $U \in \mathrm{Ult}(\mathrm{Clop}(X))$. We claim that

$$\bigcap_{C \in U} C \neq \varnothing. \tag{1.3}$$

Since U is closed under finite intersections and does not contain $\varnothing$, the family of closed sets $\{\, C \mid C \in U \,\}$ has the finite intersection property (i.e., every intersection of finitely many of its elements is nonempty). Recall that every family of closed sets with the finite intersection property has nonempty intersection. This shows (1.3).

Having established that $\bigcap_{C \in U} C \neq \varnothing$, pick x in this intersection. We claim $U = \varepsilon_X(x)$. The inclusion $U \subseteq \varepsilon_X(x)$ is obvious because for all $C \in U$ we have $x \in C$. To prove the reverse inclusion ($\varepsilon_X(x) \subseteq U$), let $D \in \mathrm{Clop}(X)$. Reasoning by contraposition, suppose $D \notin U$ and let us prove $D \notin \varepsilon_X(x)$. Since U is an ultrafilter, exactly one of D and $X \setminus D$ belongs to U; hence, $X \setminus D \in U$. Thus, $x \in X \setminus D$ (since x belongs to the intersection of all clopens in U), and so $x \notin D$, i.e. $D \notin \varepsilon_X(x)$. Thus $U = \varepsilon_X(x)$. Therefore, ε_X is surjective. ■

[a]Indeed, it is easily seen that $\varepsilon_X(x)$ is a filter. It is also clear that $\varnothing \notin \varepsilon_X(x)$. Finally, if $C \cup D \in \varepsilon_X(x)$ then $x \in C \cup D$, hence $x \in C$ or $x \in D$, i.e. $C \in \varepsilon_X(x)$ or $D \in \varepsilon_X(x)$. So $\varepsilon_X(x)$ is an ultrafilter.

[b]Indeed, every continuous bijection between compact Hausdorff spaces is a homeomorphism. This follows from the fact that every continuous map between compact Hausdorff spaces is closed, i.e. the image of a closed subset is closed. This latter fact, in turn, follows from the facts that (i) a subset of a compact Hausdorff space is closed if and only if it is compact, and (ii) every continuous map maps a compact set to a compact set.

[c]Indeed, the Stone topology on $\mathrm{Ult}(\mathrm{Clop}(X))$ has a basis of opens of the form

$$\eta_{\mathrm{Clop}(X)}(C) \;=\; \{\, U \in \mathrm{Ult}(\mathrm{Clop}(X)) \mid C \in U \,\}, \qquad (C \in \mathrm{Clop}(X)).$$

For such a basic open we have

$$\varepsilon_X^{-1}\left[\eta_{\mathrm{Clop}(X)}(C)\right] \;=\; \{\, x \in X \mid \varepsilon_X(x) \in \eta_{\mathrm{Clop}(X)}(C) \,\} \;=\; \{\, x \in X \mid C \in \varepsilon_X(x) \,\} \;=\; \{\, x \in X \mid x \in C \,\} \;=\; C,$$

which is open. Hence, ε_X is continuous.

1.6 Stone duality

We finally prove Stone duality.

Definition 1.6.1 (The categories BA and Stone)**.**

1. The category BA has Boolean algebras as objects and Boolean homomorphisms as morphisms.
2. The category Stone has Stone spaces as objects and continuous maps as morphisms.

Definition 1.6.2 (The functors Clop and Ult)**.**

1. (Clop) We define a functor
$$\mathrm{Clop}\colon \mathsf{Stone}^{\mathrm{op}} \longrightarrow \mathsf{BA}$$
as follows:
   - (On objects:) it maps a Stone space X to the Boolean algebra $\mathrm{Clop}(X)$ of clopens of X.
   - (On morphisms:) it maps a continuous map $f\colon X \to Y$ to the preimage function
$$\begin{aligned} f^{-1}[-]\colon \mathrm{Clop}(Y) &\longrightarrow \mathrm{Clop}(X) \\ A &\longmapsto f^{-1}[A]. \end{aligned}$$
2. (Ult) We define a functor
$$\mathrm{Ult}\colon \mathsf{BA} \longrightarrow \mathsf{Stone}^{\mathrm{op}}$$
as follows:
   - (On objects:) it maps a Boolean algebra A to the Stone space $\mathrm{Ult}(A)$ of ultrafilters of A.
   - (On morphisms:) it maps a Boolean homomorphism $f\colon A \to B$ to the preimage function
$$\begin{aligned} f^{-1}[-]\colon \mathrm{Ult}(B) &\longrightarrow \mathrm{Ult}(A) \\ U &\longmapsto f^{-1}[U]. \end{aligned}$$

Theorem 1.6.3 (Stone duality for Boolean algebras)**.** *The categories* BA *and* Stone *are dually equivalent*

$$\mathsf{BA} \simeq \mathsf{Stone}^{\mathrm{op}},$$

as witnessed by the functors $\mathrm{Clop}\colon \mathsf{Stone}^{\mathrm{op}} \to \mathsf{BA}$ *and* $\mathrm{Ult}\colon \mathsf{BA} \to \mathsf{Stone}^{\mathrm{op}}$ *and the natural isomorphisms* $\Big(\eta_B\colon B \to \mathrm{Clop}(\mathrm{Ult}(B))\Big)_{B\in\mathsf{BA}}$ *and* $\Big(\varepsilon_X\colon X \to \mathrm{Ult}(\mathrm{Clop}(X))\Big)_{X\in\mathsf{Stone}}$.

Proof. Consider the families of maps

$$\begin{aligned} \eta_B\colon B &\longrightarrow \mathrm{Clop}(\mathrm{Ult}(B)) \\ b &\longmapsto \{\, U \in \mathrm{Ult}(B) \mid b \in U \,\} \end{aligned} \qquad\qquad \begin{aligned} \varepsilon_X\colon X &\longrightarrow \mathrm{Ult}(\mathrm{Clop}(X)) \\ x &\longmapsto \{\, C \in \mathrm{Clop}(X) \mid x \in C \,\}. \end{aligned}$$

By the theorems proved above, each η_B is an isomorphism of Boolean algebras (Theorem 1.5.9), and each ε_X is a homeomorphism (Theorem 1.5.11).

They are natural: for every Boolean homomorphism $h\colon A \to B$ and every continuous map $f\colon X \to Y$, the diagrams

$$\begin{tikzcd} A \arrow[r, "\eta_A"] \arrow[d, "h"'] & \mathrm{Clop}(\mathrm{Ult}(A)) \arrow[d, "\mathrm{Clop}(\mathrm{Ult}(h))"] \\ B \arrow[r, "\eta_B"'] & \mathrm{Clop}(\mathrm{Ult}(B)) \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "\varepsilon_X"] \arrow[d, "f"'] & \mathrm{Ult}(\mathrm{Clop}(X)) \arrow[d, "\mathrm{Ult}(\mathrm{Clop}(f))"] \\ Y \arrow[r, "\varepsilon_Y"'] & \mathrm{Ult}(\mathrm{Clop}(Y)) \end{tikzcd}$$

commute (both are straightforward "unwind the definitions" computations, done in the footnote[a]).

Hence, Clop and Ult are quasi-inverse up to these natural isomorphisms, i.e. they define a dual equivalence.

In fact, this is an ***adjoint dual equivalence***: Clop is right adjoint[b] to Ult

$$\begin{tikzcd} \mathsf{BA} \arrow[r, shift left, "\mathrm{Ult}"] \arrow[r, phantom, "\perp"] & \mathsf{Stone}^{\mathrm{op}}, \arrow[l, shift left, "\mathrm{Clop}"] \end{tikzcd}$$

with unit $(\eta_B)_B$ and counit $(\varepsilon_X)_X$ (viewed in the opposite category); the triangle identities that one needs to verify are the following ones (for $B \in \mathsf{BA}$ and $X \in \mathsf{Stone}$, with the diagram on the left-hand side in Stone, and the diagram on the right-hand side in BA):

$$\begin{tikzcd} \mathrm{Ult}(B) \arrow[r, "\epsilon_{\mathrm{Ult}(B)}"] \arrow[dr, "\mathrm{id}_{\mathrm{Ult}(B)}"'] & \mathrm{Ult}(\mathrm{Clop}(\mathrm{Ult}(B))) \arrow[d, "\mathrm{Ult}(\eta_B)"] \\ & \mathrm{Ult}(B), \end{tikzcd} \qquad \begin{tikzcd} \mathrm{Clop}(X) \arrow[r, "\eta_{\mathrm{Clop}(X)}"] \arrow[dr, "\mathrm{id}_{\mathrm{Clop}(X)}"'] & \mathrm{Clop}(\mathrm{Ult}(\mathrm{Clop}(X))) \arrow[d, "\mathrm{Clop}(\epsilon_X)"] \\ & \mathrm{Clop}(X). \end{tikzcd}$$

The verification of the commutativity is a tedious unwinding of the definitions, done in the footnote[c]. ■

[a] For the η-square, let $a \in A$. We compute both composites as clopen subsets of $\mathrm{Ult}(B)$. For the η-square, let $a \in A$. We compute both composites as clopen subsets of $\mathrm{Ult}(B)$. First,

$$(\mathrm{Clop}(\mathrm{Ult}(h)) \circ \eta_A)(a) = \mathrm{Clop}(\mathrm{Ult}(h))\big(\eta_A(a)\big) = \mathrm{Ult}(h)^{-1}\big[\eta_A(a)\big],$$

since $\mathrm{Clop}(\mathrm{Ult}(h))$ is defined as inverse image along $\mathrm{Ult}(h)\colon \mathrm{Ult}(B) \to \mathrm{Ult}(A)$. Now let $U \in \mathrm{Ult}(B)$. Then

$$\begin{aligned} U \in \mathrm{Ult}(h)^{-1}\big[\eta_A(a)\big] &\iff \mathrm{Ult}(h)(U) \in \eta_A(a) \\ &\iff h^{-1}[U] \in \eta_A(a) \\ &\iff a \in h^{-1}[U] \\ &\iff h(a) \in U \\ &\iff U \in \eta_B(h(a)). \end{aligned}$$

Therefore $\mathrm{Ult}(h)^{-1}[\eta_A(a)] = \eta_B(h(a))$, i.e.

$$\mathrm{Clop}(\mathrm{Ult}(h))(\eta_A(a)) = \eta_B(h(a)),$$

which is exactly $(\mathrm{Clop}(\mathrm{Ult}(h)) \circ \eta_A)(a) = (\eta_B \circ h)(a)$.

For the ε-square, let $x \in X$. We compute both composites as ultrafilters on $\mathrm{Clop}(Y)$. We have

$$(\mathrm{Ult}(\mathrm{Clop}(f)) \circ \varepsilon_X)(x) = \mathrm{Ult}(\mathrm{Clop}(f))\big(\varepsilon_X(x)\big) = \mathrm{Clop}(f)^{-1}\big[\varepsilon_X(x)\big],$$

since $\mathrm{Ult}(\mathrm{Clop}(f))$ is defined by preimage along $\mathrm{Clop}(f)\colon \mathrm{Clop}(Y) \to \mathrm{Clop}(X)$. Now let $C \in \mathrm{Clop}(Y)$. Then

$$\begin{aligned} C \in \mathrm{Clop}(f)^{-1}\big[\varepsilon_X(x)\big] &\iff \mathrm{Clop}(f)(C) \in \varepsilon_X(x) \\ &\iff f^{-1}[C] \in \varepsilon_X(x) \\ &\iff x \in f^{-1}[C] \\ &\iff f(x) \in C \\ &\iff C \in \varepsilon_Y(f(x)). \end{aligned}$$

Therefore $\mathrm{Clop}(f)^{-1}[\varepsilon_X(x)] = \varepsilon_Y(f(x))$, i.e.

$$\mathrm{Ult}(\mathrm{Clop}(f))(\varepsilon_X(x)) = \varepsilon_Y(f(x)),$$

which is exactly $(\mathrm{Ult}(\mathrm{Clop}(f)) \circ \varepsilon_X)(x) = (\varepsilon_Y \circ f)(x)$.

[b]This is just because we put the "op" on Stone rather than on BA; otherwise it would have been Ult to be right adjoint to Clop.

[c]For the left triangle (in Stone), let $U \in \mathrm{Ult}(B)$. Recall that

$$\varepsilon_{\mathrm{Ult}(B)}(U) = \{\, C \in \mathrm{Clop}(\mathrm{Ult}(B)) \mid U \in C \,\} \in \mathrm{Ult}(\mathrm{Clop}(\mathrm{Ult}(B))),$$

and that $\mathrm{Ult}(\eta_B)\colon \mathrm{Ult}(\mathrm{Clop}(\mathrm{Ult}(B))) \to \mathrm{Ult}(B)$ is given by preimage:

$$\mathrm{Ult}(\eta_B)(V) = \eta_B^{-1}[V] \qquad (V \in \mathrm{Ult}(\mathrm{Clop}(\mathrm{Ult}(B)))).$$

We show $\mathrm{Ult}(\eta_B)(\varepsilon_{\mathrm{Ult}(B)}(U)) = U$ by element-chasing. For $b \in B$,

$$\begin{aligned} b \in \mathrm{Ult}(\eta_B)\big(\varepsilon_{\mathrm{Ult}(B)}(U)\big) &\iff b \in \eta_B^{-1}\big[\varepsilon_{\mathrm{Ult}(B)}(U)\big] \\ &\iff \eta_B(b) \in \varepsilon_{\mathrm{Ult}(B)}(U) \\ &\iff U \in \eta_B(b) \\ &\iff b \in U, \end{aligned}$$

hence $\mathrm{Ult}(\eta_B) \circ \varepsilon_{\mathrm{Ult}(B)} = \mathrm{id}_{\mathrm{Ult}(B)}$.

For the right triangle (in BA), let $C \in \mathrm{Clop}(X)$. Recall that

$$\eta_{\mathrm{Clop}(X)}(C) = \{\mathcal{U} \in \mathrm{Ult}(\mathrm{Clop}(X)) \mid C \in \mathcal{U}\} \in \mathrm{Clop}(\mathrm{Ult}(\mathrm{Clop}(X))),$$

and that $\mathrm{Clop}(\varepsilon_X)\colon \mathrm{Clop}(\mathrm{Ult}(\mathrm{Clop}(X))) \to \mathrm{Clop}(X)$ is inverse image along ε_X:

$$\mathrm{Clop}(\varepsilon_X)(D) = \varepsilon_X^{-1}[D] \qquad (D \in \mathrm{Clop}(\mathrm{Ult}(\mathrm{Clop}(X)))).$$

Then for $x \in X$,

$$\begin{aligned} x \in \mathrm{Clop}(\varepsilon_X)\big(\eta_{\mathrm{Clop}(X)}(C)\big) &\iff \varepsilon_X(x) \in \eta_{\mathrm{Clop}(X)}(C) \\ &\iff C \in \varepsilon_X(x) \\ &\iff x \in C, \end{aligned}$$

so $\mathrm{Clop}(\varepsilon_X) \circ \eta_{\mathrm{Clop}(X)} = \mathrm{id}_{\mathrm{Clop}(X)}$.

1.7 Using Stone duality

In this section, we collect some simple examples of the use of Stone duality.

1.7.1 Spaces of models and the compactness theorem

Remark 1.7.1. In Proposition 1.3.2 we stated that, for every propositional theory $\mathcal{T}$ in a propositional language $\mathcal{L}$, there is a bijection

$$\mathrm{Mod}(\mathcal{T}) \longrightarrow \hom(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}, 2).$$

Composing this bijection with the bijection

$$\hom(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}, 2) \longrightarrow \mathrm{Ult}(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}),$$

we obtain the following bijection:

$$\begin{aligned} \mathrm{Mod}(\mathcal{T}) &\longrightarrow \mathrm{Ult}(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}) \\ v &\longmapsto \big\{\, [\,\varphi\,] \in \mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}} \mid v \models \varphi \,\big\}. \end{aligned} \tag{1.4}$$

Via this bijection, $\mathrm{Mod}(\mathcal{T})$ inherits a Stone topology from $\mathrm{Ult}(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}})$. This can be defined directly as follows.

Definition 1.7.2 (The space of models)**.** Let $\mathcal{T}$ be a propositional theory in a propositional language $\mathcal{L}$. We equip $\mathrm{Mod}(\mathcal{T})$ with the topology that has as a basis the sets of the form

$$[\![\varphi]\!]_{\mathcal{T}} \;:=\; \{\, v \in \mathrm{Mod}(\mathcal{T}) \mid v \models \varphi \,\}, \qquad\qquad \varphi \in \mathrm{Form}(\mathcal{L}).$$

We now characterize the clopens.

Proposition 1.7.3 (The space of models is a Stone space)**.** $\mathrm{Mod}(\mathcal{T})$ *is a Stone space, whose clopens are precisely the sets of the form*

$$[\![\varphi]\!]_{\mathcal{T}} \;=\; \{\, v \in \mathrm{Mod}(\mathcal{T}) \mid v \models \varphi \,\}, \qquad\qquad \varphi \in \mathrm{Form}(\mathcal{L}).$$

Proof. It is enough to observe that, under the bijection (1.4) in Remark 1.7.1, the subset $[\![\varphi]\!]$ corresponds to the basic open set $\eta_{\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}}([\,\varphi\,])$ of $\mathrm{Ult}(\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}})$, and that $\mathrm{Ult}(\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}})$ is a Stone space with clopens exactly the sets of the form $\eta_{\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}}([\,\varphi\,])$ for $\varphi \in \mathrm{Form}(\mathcal{L})$. ■

Proposition 1.7.4 (No propositional theory captures "exactly one among countably many")**.** *Let* $\mathcal{L} = \{p_0, p_1, p_2, \dots\}$*. There is no propositional theory* $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ *such that* $\mathrm{Mod}(\mathcal{T})$ *consists exactly of those valuations* $v \colon \mathcal{L} \to 2$ *for which there is* exactly one $n \in \mathbb{N}$ *with* $v(p_n) = 1$*.*

Proof. Suppose, for a contradiction, that such a theory $\mathcal{T}$ exists. For each $n \in \mathbb{N}$ consider the set

$$U_n \;:=\; [\![p_n]\!]_{\mathcal{T}} \;=\; \{\, v \in \mathrm{Mod}(\mathcal{T}) \mid v \models p_n \,\}.$$

Each U_n is clopen in $\mathrm{Mod}(\mathcal{T})$ by Proposition 1.7.3, and by assumption we have a cover

$$\mathrm{Mod}(\mathcal{T}) = \bigcup_{n \in \mathbb{N}} U_n.$$

Since $\mathrm{Mod}(\mathcal{T})$ is compact, there is a finite $I \subseteq \mathbb{N}$ such that $\mathrm{Mod}(\mathcal{T}) = \bigcup_{n \in I} U_n$.

Choose $m \in \mathbb{N} \setminus I$, and define the valuation $v_m \colon \mathcal{L} \to 2$ by $v_m(p_m) = 1$ and $v_m(p_n) = 0$ for $n \neq m$. By the defining property of $\mathcal{T}$, the valuation v_m belongs to $\mathrm{Mod}(\mathcal{T})$. But $v_m \notin U_n$ for every $n \in I$ (since $v_m(p_n) = 0$), contradicting $\mathrm{Mod}(\mathcal{T}) = \bigcup_{n \in I} U_n$. ■

Definition 1.7.5 (Semantic consequence: "theory $\models$ formula")**.** Let $\mathcal{T}$ be a propositional theory in a propositional language $\mathcal{L}$, and let $\varphi \in \mathrm{Form}(\mathcal{L})$. We write

$$\mathcal{T} \models \varphi$$

and say that φ ***is a semantic consequence of*** $\mathcal{T}$ if every model of $\mathcal{T}$ satisfies φ.

(Yes, we use the same symbol used for the relation of satisfiability between valuations and formulas.)

The compactness of the space of ultrafilters has the following logical flavour.

Theorem 1.7.6 (Compactness theorem of classical propositional logic)**.** *Let* $\mathcal{L}$ *be a propositional language, let* $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ *be a propositional theory, and let* $\varphi \in \mathrm{Form}(\mathcal{L})$*. If* $\mathcal{T} \models \varphi$*, then there is a finite* $\mathcal{T}_0 \subseteq \mathcal{T}$ *such that* $\mathcal{T}_0 \models \varphi$*.*

Proof. Consider the Stone space $\mathrm{Mod}(\varnothing)$ (the space of all valuations on $\mathcal{L}$). For each formula $\psi \in \mathrm{Form}(\mathcal{L})$ set

$$[\![\psi]\!] \;:=\; \{\, v \in \mathrm{Mod}(\varnothing) \mid v \models \psi \,\}.$$

By Proposition 1.7.3 (with $\mathcal{T} = \varnothing$), each $[\![\psi]\!]$ is clopen, and hence closed.

The assumption says exactly that

$$\bigcap_{\sigma \in \mathcal{T}} [\![\sigma]\!] \;\subseteq\; [\![\varphi]\!].$$

From the definition of compactness, it follows immediately that, if the intersection of a family of closed subsets is contained in an open set, then there is a *finite* subfamily whose intersection is contained in the open set. Since every $[\![\sigma]\!]$ (for $\sigma \in \mathcal{T}$) is clopen and hence closed, and $[\![\varphi]\!]$ is clopen and hence open, there are $\sigma_1, \ldots, \sigma_n \in \mathcal{T}$ such that

$$[\![\sigma_1]\!] \cap \cdots \cap [\![\sigma_n]\!] \subseteq [\![\varphi]\!].$$

Let $\mathcal{T}_0 := \{\sigma_1, \ldots, \sigma_n\}$. This is exactly the desired conclusion. ■

1.7.2 Question: are coproduct maps injective?

Question (Guiding question)**.** *Let A and B be Boolean algebras. Is the coproduct morphism*

$$i_A \colon A \longrightarrow A + B$$

injective?

In logical terms, suppose A and B are Lindenbaum–Tarski algebras of theories $\mathcal{T}_A$ and $\mathcal{T}_B$ in disjoint languages $\mathcal{L}_A$ and $\mathcal{L}_B$. One can easily show that the injectivity of a Boolean homomorphism between Boolean algebras is equivalent to the condition that a non-1 element is mapped to a non-1 element. The question then translates to the following:

is it true that, for every formula φ in the language $\mathcal{L}_A$,
if φ does not follow from $\mathcal{T}_A$ (i.e., $\mathcal{T}_A \not\models \varphi$),
then φ does not follow from $\mathcal{T}_A \sqcup \mathcal{T}_B$ (i.e., $\mathcal{T}_A \sqcup \mathcal{T}_B \not\models \varphi$)?

We answer this by translating the question to Stone via Stone duality. To do so, we make a brief detour in which

- we show that binary products of Stone spaces (and more generally, all limits) are easy: they are computed as in Set,
- we show that the injectivity (resp. surjectivity) of a Boolean homomorphism is equivalent to the surjectivity (resp. injectivity) of the dual continuous map.

1.7.3 Limits and colimits in BA and Stone

Limits and colimits in BA

The category of Boolean algebras is complete and cocomplete, as any equationally defined class of algebras. However, there is a fundamental difference in the behaviour of limits and colimits:

- (Limits ✓) Limits of Boolean algebras are great: they are computed as in Set, i.e., the forgetful functor $\mathsf{BA} \to \mathsf{Set}$ preserves limits (as the forgetful functor of any equationally defined class of algebras).
- (Colimits ✗) However, colimits are crazy, meaning that they are not at all computed as in Set. For example:
  - the initial object of BA is 2 (while in Set the initial object is the empty set).
  - In BA, $2 + 2 = 2$ (since 2 is the initial object), while in Set the coproduct of a 2-element set with itself has 4 elements.

Limits and colimits in Stone

Since Stone is dually equivalent to BA, which is complete and cocomplete since BA is an equationally defined class of algebras, Stone is also complete and cocomplete.

In Stone, limits are nice (= computed as in Set):

Lemma 1.7.7 (Closed in Stone is Stone)**.** *Every closed subspace of a Stone space is a Stone space (with the subspace topology).*

Proof. Let X be a Stone space and let $F \subseteq X$ be closed. Since F is closed in the compact space X, F is compact, as well. To see that F is totally separated, let $x \neq y$ in F. Since X is totally separated, there is $C \in \mathrm{Clop}(X)$ such that $x \in C$ and $y \notin C$. Then $C \cap F$ is clopen in F and separates x and y. ∎

Proposition 1.7.8 (Limits in Stone)**.** *The category* Stone *is complete and the forgetful functor*

$$\mathsf{Stone} \to \mathsf{Set}$$

preserves limits.
In fact, limits in Stone *are computed as in* Top*:*

1. *(Products) the product in* Stone *of a family* $(X_i)_{i \in I}$ *of objects of* Stone *is the Cartesian product* $\prod_{i \in I} X_i$ *with the product topology (and the projection as limit morphisms);*
2. *(Equalizer) the equalizer in* Stone *of a pair of morphisms* $f, g \colon X \to Y$ *in* Stone *is the space*

$$\{\, x \in X \mid f(x) = g(x) \,\},$$

with the subspace topology (and the inclusion as limit morphism).

Proof. It is enough to prove that Stone is closed in Top under limits. (We recall that the forgetful functor Top $\to$ Set preserves limits.)

(1). We prove that the topological product $\prod_{i \in I} X_i$ of Stone spaces is a Stone space.

- (Compactness) We recall that Tychonoff's theorem states that an arbitrary product of compact spaces is compact. Therefore, an arbitrary product of Stone spaces is compact.[a]
- (Total separation) Let us prove that $\prod_{i \in I} X_i$ is totally separated. Let $(x_j)_j$ and $(y_j)_j$ be distinct elements. Then, they differ at some component, i.e., there is $i \in I$ such that $x_i \neq y_i$. Then, there is a clopen C of X_i such that $x_i \in C$ and $y_i \notin C$. Then, letting $\pi_i \colon \prod_j X_j \to X_i$ denote the i-th projection, the set $\pi_i^{-1}[C]$ is a clopen subset of $\prod_j X_j$ in the product topology, and $(x_j)_j \in \pi_i^{-1}[C]$ and $(y_j)_j \notin \pi_i^{-1}[C]$. Thus, $\prod_{i \in I} X_i$ is totally separated.

(2). We prove that $\{\, x \in X \mid f(x) = g(x) \,\}$ is a Stone space. Consider the map $\langle f, g \rangle \colon X \to Y \times Y$, $\langle f, g \rangle(x) = (f(x), g(x))$. Then $\mathsf{Eq}(f, g) = \langle f, g \rangle^{-1}[\Delta_Y]$, where $\Delta_Y = \{(y, y) \mid y \in Y\}$ is the diagonal. Since Y is Hausdorff, Δ_Y is closed in $Y \times Y$, and hence $\{\, x \in X \mid f(x) = g(x) \,\}$ is closed in X. Since every closed subspace of a Stone space is a Stone space (Lemma 1.7.7), $\{\, x \in X \mid f(x) = g(x) \,\}$ is a Stone space, as desired. ∎

[a] We recall that Tychonoff's theorem, stated in its full generality (for arbitrary compact spaces), is equivalent to the Axiom of Choice.

Remark 1.7.9 (Colimits in Stone are not as in Set)**.** While Stone is cocomplete (since BA is complete), colimits are, in general, not computed as in Set. A simple example of a colimit that is *not* preserved by

the forgetful functor $\mathsf{Stone} \to \mathsf{Set}$ is the coproduct of countably many copies of a singleton. A proof is in the footnote; the idea is that one would like to equip $\mathbb{N}$ with the discrete topology, which, however, is not compact.[8] By Stone duality, the coproduct $\sum_{n\in\mathbb{N}} 1$ in Stone exists and is the Stone space $\mathrm{Ult}(\prod_{n\in\mathbb{N}} \mathrm{Clop}(1)) = \mathrm{Ult}(\prod_{n\in\mathbb{N}} 2) = \mathrm{Ult}(\mathcal{P}(X))$.

In general, infinite coproducts, coequalizers and pushouts are not preserved by the forgetful functor to Set. See the footnote for the case of coequalizers and pushouts.[9]

Finite coproducts, instead, *are* computed as in Set, and in fact as in Top: a finite coproduct of Stone spaces is the disjoint union equipped with the disjoint union topology.

In summary:

	BA	Stone
Has all limits / colimits?	✓/ ✓	✓/ ✓
Are limits computed as in Set?	✓	✓
Are colimits computed as in Set?	✗	✗ (exception: *finite coproducts* are as in Set)

[8]We prove that the forgetful functor $U\colon \mathsf{Stone} \to \mathsf{Set}$ does *not* preserve the coproduct of countably many copies of the one-point Stone space.

Suppose it does, and let us reach a contradiction. Let $(\iota_n\colon 1_n \to C)_{n\in\mathbb{N}}$ be a coproduct cocone in Stone. Then $(U(\iota_n))_{n\in\mathbb{N}}$ is a coproduct cocone in Set, hence it induces a canonical bijection

$$\mathbb{N} \cong U(C),$$

under which $U(\iota_n)$ picks out the element $n \in U(C)$. Transporting the topology of C along this bijection, we may assume $C = (\mathbb{N}, \tau)$ for some topology τ.

We claim that τ is discrete. Let $A \subseteq \mathbb{N}$ be arbitrary, and consider the discrete two-point Stone space $2 = \{0, 1\}$. Define, for each $n \in \mathbb{N}$, a (necessarily continuous) map

$$G_n\colon 1_n \to 2, \qquad G_n(*) = \begin{cases} 1 & \text{if } n \in A, \\ 0 & \text{if } n \notin A. \end{cases}$$

By the universal property of the coproduct $C = \sum_{n\in\mathbb{N}} 1_n$ in Stone, there is a unique continuous map $f_A\colon (\mathbb{N}, \tau) \to 2$ such that $f_A \circ \iota_n = G_n$ for all $n \in \mathbb{N}$. Since $U(\iota_n)$ picks out $n \in \mathbb{N}$, this forces $f_A(n) = G_n(*)$ for all n, hence $f_A^{-1}[\{1\}] = A$. As $\{1\}$ is clopen in 2 and f_A is continuous, it follows that A is clopen in $(\mathbb{N}, \tau)$. Since A was arbitrary, every subset of $\mathbb{N}$ is clopen, so τ is discrete. But an infinite discrete space is not compact, whereas every Stone space is compact, a contradiction.

[9]Let $U\colon \mathsf{Stone} \to \mathsf{Set}$ be the forgetful functor. We sketch a proof of the fact that U does *not* preserve coequalizers. (The analogous statement for pushouts follows, since U preserves the initial object.)

Let $f\colon B \to [0, 1]$ be a continuous surjection, where B is a Stone space and $[0, 1]$ carries the usual Euclidean topology. (Such an f exists: for instance, take $B = \beta([0, 1]_{\mathrm{disc}}) = \mathrm{Ult}(\mathcal{P}([0, 1]))$ and extend the identity map $[0, 1]_{\mathrm{disc}} \to [0, 1]$ along the Stone–Čech compactification; by the universal property of β, the extension exists and is surjective.)

Consider the *kernel pair* of f in Top (equivalently in $\mathsf{CompHaus}$): $R := \{(x, y) \in B \times B \mid f(x) = f(y)\}$. Since $[0, 1]$ is Hausdorff, the diagonal $\Delta \subseteq [0, 1] \times [0, 1]$ is closed, and $R = (f \times f)^{-1}[\Delta]$ is a closed subset of $B \times B$. As B is Stone, so is $B \times B$, and every closed subspace of a Stone space is Stone. Hence, R is a Stone space. Let $r_1, r_2\colon R \rightrightarrows B$ be the two projections.

Claim. The coequalizer of r_1, r_2 in Stone is the one-point Stone space 1.

Indeed, the constant map $q\colon B \to 1$ clearly coequalizes r_1, r_2. Now let $g\colon B \to X$ be a continuous map into a Stone space such that $g \circ r_1 = g \circ r_2$. Then $g(x) = g(y)$ whenever $f(x) = f(y)$, i.e. g is constant on each fiber of f. Define

$$h\colon [0, 1] \to X, \qquad h(t) := g(x) \ \text{ for any } x \in f^{-1}(t).$$

This is well-defined by the fiber-constancy of g, and it satisfies $g = h \circ f$.

Because B is compact and $[0, 1]$ is Hausdorff, the map f is a quotient map; hence from the continuity of $g = h \circ f$ it follows that h is continuous. Since $[0, 1]$ is connected, the image $h([0, 1])$ is connected. But every connected subset of a Stone space is a singleton (Stone spaces are totally disconnected). Therefore, h is constant, and so is $g = h \circ f$. Consequently, g factors uniquely through $q\colon B \to 1$. This proves that $\mathrm{coeq}_{\mathsf{Stone}}(r_1, r_2) \cong 1$.

Conclusion. Applying U, we have $U(\mathrm{coeq}_{\mathsf{Stone}}(r_1, r_2)) = U(1)$, a singleton. On the other hand, in Set the coequalizer of $U(r_1)$ and $U(r_2)$ is the quotient of the set $U(B)$ by the equivalence relation generated by $x \sim y \iff f(x) = f(y)$; since f is surjective, this quotient is (canonically) in bijection with $[0, 1]$, hence it is not a singleton. Therefore, U does not preserve this coequalizer, and thus the forgetful functor $\mathsf{Stone} \to \mathsf{Set}$ does not preserve coequalizers.

1.7.4 Injective is dual to surjective, and vice versa

Lemma 1.7.10 (Stone spaces have a basis of clopens)**.** *Every Stone space is* 0**-dimensional**, *i.e., it has a basis of clopens. In other words, for every open U and $x \in U$, there is a clopen C such that $x \in C \subseteq U$.*

Proof. Let X be a Stone space, U an open of X, and $x \in U$. Set $F \coloneqq X \setminus U$, which is closed. For each $y \in F$ we have $y \neq x$, so by total separation there is a clopen $C_y \subseteq X$ with

$$x \in C_y \quad \text{and} \quad y \notin C_y.$$

Equivalently, $y \in X \setminus C_y$, and the latter is open. Thus the family $\{ X \setminus C_y \mid y \in F \}$ is an open cover of F.

Since F is a closed subspace of the compact space X, it is compact. Hence, there are $y_1, \dots, y_n \in F$ such that

$$F \subseteq (X \setminus C_{y_1}) \cup \cdots \cup (X \setminus C_{y_n}).$$

Let

$$C \coloneqq C_{y_1} \cap \cdots \cap C_{y_n}.$$

Then C is clopen, $x \in C$, and moreover $C \cap F = \varnothing$ (because F is contained in the union of the complements). Therefore, $C \subseteq X \setminus F = U$, as desired. ■

Lemma 1.7.11 (Clopens separate disjoint closed sets)**.** *Let X be a Stone space and let $A, B \subseteq X$ be closed and disjoint. There is a clopen C such that $A \subseteq C$ and $B \cap C = \varnothing$.*

Proof. For each $a \in A$, the set $X \setminus B$ is an open neighbourhood of a. Since X has a basis of clopens (Lemma 1.7.10), there is $C_a \in \mathrm{Clop}(X)$ with $a \in C_a \subseteq X \setminus B$. Then $(C_a)_{a \in A}$ is an open cover of A. As A is closed in the compact space X, it is compact. Therefore, there are $a_1, \dots, a_n \in A$ such that $A \subseteq C_{a_1} \cup \cdots \cup C_{a_n}$. Set $C \coloneqq C_{a_1} \cup \cdots \cup C_{a_n}$. This is clopen, contains A, and is disjoint from B. ■

Theorem 1.7.12 (Injective is dual to surjective, and vice versa)**.** *Let $f \colon X \to Y$ be a continuous map of Stone spaces. Then:*

1. *f is injective iff $f^{-1}[-] \colon \mathrm{Clop}(Y) \to \mathrm{Clop}(X)$ is surjective;*
2. *f is surjective iff $f^{-1}[-] \colon \mathrm{Clop}(Y) \to \mathrm{Clop}(X)$ is injective.*

Proof. (1). ($\Rightarrow$). Suppose f is injective and let $A \in \mathrm{Clop}(X)$. Since X is compact and Y is Hausdorff, f is a homeomorphism onto the compact subset $f[X] \subseteq Y$, which is therefore closed. The sets $f[A]$ and $f[X \setminus A]$ are compact, hence closed in Y, and they are disjoint. By Lemma 1.7.11, there is $C \in \mathrm{Clop}(Y)$ such that $f[A] \subseteq C$ and $f[X \setminus A] \cap C = \varnothing$. Then $f^{-1}[C] = A$, so $f^{-1}[-]$ is surjective.

($\Leftarrow$). Conversely, suppose f is not injective and let us prove that $f^{-1}[-]$ is not surjective. Pick $x \neq y$ in X with $f(x) = f(y)$. Since X is totally separated, choose $A \in \mathrm{Clop}(X)$ with $x \in A$ and $y \notin A$. There is no clopen C of Y such that $f^{-1}[C] = A$, because for every clopen C of Y we have $x \in f^{-1}[C] \Leftrightarrow y \in f^{-1}[C]$; therefore, $f^{-1}[-]$ is not surjective. Hence, (2). ($\Rightarrow$). Suppose that f is surjective. To prove injectivity of $f^{-1}[-]$, suppose $f^{-1}[C] = f^{-1}[D]$ for clopens $C, D \subseteq Y$, and let us prove that $C = D$. Let $y \in Y$ and let us prove that $y \in C \iff y \in D$. By surjectivity, there is x with $f(x) = y$. Thus, $y \in C \Leftrightarrow x \in f^{-1}[C] = f^{-1}[D] \Leftrightarrow y \in D$. Hence, $C = D$, so $f^{-1}[-]$ is injective.

($\Leftarrow$). Conversely, suppose that f is not surjective, i.e., there is $y_0 \in Y \setminus f[X]$. Since $f[X]$ and $\{y_0\}$ are disjoint closed sets, by Lemma 1.7.11 there is $C \in \mathrm{Clop}(Y)$ such that $f[X] \subseteq C$ and $y_0 \notin C$. Then $f^{-1}[C] = X = f^{-1}[Y]$ but $C \neq Y$, so $f^{-1}[-]$ is not injective. ■

1.7.5 Answer to the question "are coproduct maps injective?"

We are now ready to answer our guiding question:

Question 1.7.13. (Guiding question) Let A and B be Boolean algebras. Is the coproduct morphism

$$i_A \colon A \longrightarrow A + B$$

injective?

Answer 1.7.14 (Answer to Question 1.7.13)**.** The coproduct morphism

$$i_A \colon A \longrightarrow A + B$$

is injective if and only if either A is trivial (i.e., the one-element Boolean algebra) or B is nontrivial. Equivalently, i_A fails to be injective if and only if A is nontrivial and B is trivial.

Proof. Let

$$X \coloneqq \mathrm{Ult}(A) \qquad \text{and} \qquad Y \coloneqq \mathrm{Ult}(B).$$

Under Stone duality, coproducts in BA correspond to products in Stone, and the morphism

$$i_A \colon A \longrightarrow A + B$$

is dual to the first projection

$$\pi_X \colon X \times Y \longrightarrow X.$$

By Theorem 1.7.12, i_A is injective if and only if π_X is surjective.

Now, π_X is surjective if and only if either $Y \neq \varnothing$ or $X = \varnothing$. Indeed, if $Y \neq \varnothing$, choose $y_0 \in Y$. Then for every $x \in X$ we have $\pi_X(x, y_0) = x$, so π_X is surjective. On the other hand, if $Y = \varnothing$, then $X \times Y = \varnothing$, so π_X is the unique map

$$\varnothing \longrightarrow X,$$

which is surjective if and only if $X = \varnothing$.

Finally, $X = \varnothing$ if and only if A is trivial, and $Y = \varnothing$ if and only if B is trivial. Therefore, i_A is injective if and only if either A is trivial or B is nontrivial. ∎

In logical terms, suppose A and B are Lindenbaum–Tarski algebras of theories $\mathcal{T}_A$ and $\mathcal{T}_B$ in disjoint languages $\mathcal{L}_A$ and $\mathcal{L}_B$. One can easily show that the injectivity of a Boolean homomorphism between Boolean algebras is equivalent to the condition that a non-1 element is mapped to a non-1 element. The question then translates to the following:

is it true that, for every formula φ in the language $\mathcal{L}_A$,
if φ does not follow from $\mathcal{T}_A$ (i.e., $\mathcal{T}_A \not\models \varphi$),
then φ does not follow from $\mathcal{T}_A \sqcup \mathcal{T}_B$ (i.e., $\mathcal{T}_A \sqcup \mathcal{T}_B \not\models \varphi$)?

Answer 1.7.14 tells us that this is always true, except when $\mathcal{T}_B$ is inconsistent (i.e., $\mathcal{T}_B \models 0$) and $\mathcal{T}_A$ is not (i.e., $\mathcal{T}_A \not\models 0$). Intuitively, if $\mathcal{T}_B$ is inconsistent and $\mathcal{T}_A$ is not, then the theory $\mathcal{T}_A \sqcup \mathcal{T}_B$ proves the falsum 0. Since 0 is also a formula of $\mathcal{L}_A$, this gives a formula of the language $\mathcal{L}_A$ that was not provable from $\mathcal{T}_A$ but becomes provable after adding $\mathcal{T}_B$.

1.7.6 Monomorphisms and epimorphisms in BA and Stone

We now summarize how the categorical notions of monomorphism, epimorphism, regular monomorphism and regular epimorphism are characterized in BA and in Stone. To prove some of these characterizations, we use Stone duality: we characterize monos and regular epis in each category, and then use Stone duality to characterize epis and regular monos on the dual category.

Theorem 1.7.15 (Monos, regular monos, epis and regular epis in BA and Stone)**.**

1. *In* BA*, monomorphisms and regular monomorphisms are precisely the injective homomorphisms.*
2. *In* BA*, epimorphisms and regular epimorphisms are precisely the surjective homomorphisms.*
3. *In* Stone*, monomorphisms and regular monomorphisms are precisely the injective continuous maps.*
4. *In* Stone*, epimorphisms and regular epimorphisms are precisely the surjective continuous maps.*

Proof. (Monos and regular epis in BA, epis and regular monos in Stone:) Recall that BA is a variety of algebras, i.e., an equationally defined class of algebras. In any variety of algebras, the monomorphisms are precisely the injective homomorphisms, and the regular epimorphisms are precisely the surjective homomorphisms. Therefore, the monomorphisms in BA are precisely the injective homomorphisms, and the regular epimorphisms in BA are precisely the surjective homomorphisms. By Stone duality and Theorem 1.7.12, we immediately obtain that, in Stone, the epimorphisms are precisely the surjective continuous maps, and the regular monomorphisms are precisely the injective continuous maps.

(Epis in BA, monos in Stone:) Proving directly that epimorphisms in BA are surjective is not immediate. To simplify the argument, we use Stone duality. Let us prove that the monomorphisms in Stone are precisely the injective continuous maps.

- First, the easiest direction: every injective morphism in Stone is a monomorphism, since this is true in Set and faithful functors reflect monomorphisms.
- For the converse direction, let $f\colon X \to Y$ be a continuous map that is not injective, and let us prove that it is not a monomorphism. Pick $x_1 \neq x_2$ in X such that $f(x_1) = f(x_2)$. Let 1 be the one-point space, which is a Stone space, and define
$$g_1, g_2\colon 1 \longrightarrow X$$
to be the constant maps with values x_1 and x_2, respectively. Then $g_1 \neq g_2$, but
$$f \circ g_1 = f \circ g_2,$$
so f is not a monomorphism.

Therefore, monomorphisms in Stone are precisely the injective continuous maps. By Stone duality and Theorem 1.7.12 (injective is dual to surjective, and vice versa), it follows that the epimorphisms in BA are precisely the surjective homomorphisms.

(Regular monos in BA, regular epis in Stone:) Also proving that injective homomorphisms in BA are regular monomorphisms is not immediate. The obvious idea would be to take the equalizer of the *cokernel pair* of one such injective morphism, but the cokernel pair in BA is not computed as in Set, and is thus difficult to manage. So, we use Stone duality: we characterize regular epimorphisms in Stone as precisely the surjective continuous maps.

- First, the easiest direction: since every regular epimorphism is an epimorphism, and epimorphisms in Stone are surjective, it follows that regular epimorphisms in Stone are surjective continuous maps.
- Let us prove the converse implication: let $f\colon X \to Y$ be a surjective continuous map of Stone spaces, and let us prove that it is a regular epimorphism. Consider its kernel pair
$$R \coloneqq \{(x_1, x_2) \in X \times X \mid f(x_1) = f(x_2)\},$$
with the two projections
$$r_1, r_2\colon R \rightrightarrows X.$$

Since Y is Hausdorff, the diagonal Δ_Y of $Y \times Y$ is closed, and hence the set $R = (f \times f)^{-1}[\Delta_Y]$ is a closed subspace of $X \times X$. As $X \times X$ is a Stone space, so is R.

Now let $g\colon X \to Z$ be a continuous map such that

$$g \circ r_1 = g \circ r_2.$$

Since surjections are coequalizers of their kernel pairs in Set, there is a unique set-theoretic map $\bar{g}\colon Y \to Z$ making the triangle in the following diagram commute.

$$\begin{tikzcd} R \arrow[r, shift left, "r_1"] \arrow[r, shift right, "r_2"'] & X \arrow[r, "f"] \arrow[dr, "g"'] & Y \arrow[d, dashed, "\bar{g}"] \\ & & Z \end{tikzcd}$$

We claim that $\bar{g}$ is continuous. To prove continuity of $\bar{g}$, it is enough to show that the preimage of every closed subset of Z is closed in Y. Let $C \subseteq Z$ be closed. Then

$$\bar{g}^{-1}[C] = f[g^{-1}[C]].$$

Indeed, if $y \in Y$, choose $x \in X$ such that $f(x) = y$. Then

$$y \in \bar{g}^{-1}[C] \;\Leftrightarrow\; \bar{g}(y) \in C \;\Leftrightarrow\; g(x) \in C \;\Leftrightarrow\; x \in g^{-1}[C] \;\Leftrightarrow\; y \in f[g^{-1}[C]].$$

Now $g^{-1}[C]$ is closed in the compact space X, hence compact. Its image under the continuous map f is compact, and therefore closed in the Hausdorff space Y. So $\bar{g}^{-1}[C]$ is closed, which proves that $\bar{g}$ is continuous. Hence, f is the coequalizer of r_1, r_2 in Stone, and therefore f is a regular epimorphism.

Therefore, regular epimorphisms in Stone are precisely the surjective continuous maps. By Stone duality and Theorem 1.7.12 (injective is dual to surjective, and vice versa), it follows that the regular monomorphisms in BA are precisely the injective homomorphisms. ■

Remark 1.7.16. The fact that epimorphisms in BA are surjective has a logical counterpart: it is related to Beth definability.

1.7.7 Amalgamation

We conclude this part with another useful categorical property of BA.

Theorem 1.7.17 (Amalgamation property of Boolean algebras)**.** *Given injective Boolean homomorphisms*

$$\begin{tikzcd} A \arrow[r, hook] \arrow[d, hook] & B \\ C & \end{tikzcd}$$

there are a Boolean algebra D and injective homomorphisms

$$\begin{tikzcd} A \arrow[r, hook] \arrow[d, hook] & B \arrow[d, hook] \\ C \arrow[r, hook] & D \end{tikzcd}$$

such that the square commutes.

Proof. Let

$$i\colon A \hookrightarrow B \qquad \text{and} \qquad j\colon A \hookrightarrow C$$

be injective homomorphisms. By Stone duality and Theorem 1.7.12 ("Injective is dual to surjective, and vice versa"), they correspond to surjective continuous maps

$$\mathrm{Ult}(B) \twoheadrightarrow \mathrm{Ult}(A) \qquad \text{and} \qquad \mathrm{Ult}(C) \twoheadrightarrow \mathrm{Ult}(A).$$

Let

$$P := \mathrm{Ult}(B) \times_{\mathrm{Ult}(A)} \mathrm{Ult}(C)$$

be their pullback in Stone. By Proposition 1.7.8, it is computed as in Set, so

$$P = \{(x, y) \in \mathrm{Ult}(B) \times \mathrm{Ult}(C) \mid f(x) = g(y)\},$$

where

$$f\colon \mathrm{Ult}(B) \to \mathrm{Ult}(A) \qquad \text{and} \qquad g\colon \mathrm{Ult}(C) \to \mathrm{Ult}(A)$$

are the two surjective maps. Thus, we have a commutative diagram

$$\begin{tikzcd} P \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & \mathrm{Ult}(C) \arrow[d, "g"] \\ \mathrm{Ult}(B) \arrow[r, "f"'] & \mathrm{Ult}(A). \end{tikzcd}$$

We claim that both projections π_1 and π_2 are surjective. We prove this for π_1; the proof for π_2 is symmetric. Let $x \in \mathrm{Ult}(B)$. Since g is surjective, there is $y \in \mathrm{Ult}(C)$ such that

$$g(y) = f(x).$$

Then $(x, y) \in P$, and therefore

$$\pi_1(x, y) = x.$$

So π_1 is surjective.

Now dualize the above pullback square. Since surjective maps in Stone correspond to injective homomorphisms in BA, we obtain a commutative square

$$\begin{tikzcd} A \arrow[r, hook] \arrow[d, hook] & B \arrow[d, hook] \\ C \arrow[r, hook] & \mathrm{Clop}(P), \end{tikzcd}$$

which proves the claim. ∎

Remark 1.7.18. The amalgamation property of BA has a logical counterpart: it corresponds to a property called interpolation.

1.7.8 Free Boolean algebras and counting formulas modulo equivalence

We now settle the question: given a finite language $\mathcal{L}$ of cardinality n, how many formulas (up to semantic equivalence) are there?

Theorem 1.7.19 (Formulas modulo equivalence are free Boolean algebras)**.** *For every propositional language $\mathcal{L}$, the Boolean algebra* $\mathrm{Form}(\mathcal{L})/\equiv$ *is the free Boolean algebra on $\mathcal{L}$.*

Proof. Let B be a Boolean algebra, and let

$$v\colon \mathcal{L} \longrightarrow B$$

be a function.

Since $\mathrm{Form}(\mathcal{L})$ is the algebra of formulas built from the generators in $\mathcal{L}$ using the Boolean connectives, there is a unique homomorphism

$$\bar{v}\colon \mathrm{Form}(\mathcal{L}) \longrightarrow B$$

such that the following triangle commutes:

$$\begin{tikzcd} \mathcal{L} \arrow[r, hook] \arrow[dr, "v"'] & \mathrm{Form}(\mathcal{L}) \arrow[d, "\bar{v}"] \\ & B. \end{tikzcd}$$

Now let

$$\pi\colon \mathrm{Form}(\mathcal{L}) \twoheadrightarrow \mathrm{Form}(\mathcal{L})/{\equiv}$$

be the quotient map. To prove that $\mathrm{Form}(\mathcal{L})/{\equiv}$ is the free Boolean algebra on $\mathcal{L}$, it is enough to show that $\bar{v}$ factors uniquely through π, i.e., that there is a unique Boolean homomorphism

$$\widetilde{v}\colon \mathrm{Form}(\mathcal{L})/{\equiv} \longrightarrow B$$

making the following diagram commute:

$$\begin{tikzcd} \mathcal{L} \arrow[r] \arrow[dr, "v"'] & \mathrm{Form}(\mathcal{L}) \arrow[r, two heads, "\pi"] \arrow[d, "\bar{v}"] & \mathrm{Form}(\mathcal{L})/{\equiv} \arrow[dl, dashed, "\widetilde{v}"] \\ & B & \end{tikzcd}.$$

So it remains to show that $\bar{v}$ is constant on $\equiv$-equivalence classes. In other words, whenever $\varphi \equiv \psi$, we must prove that

$$\bar{v}(\varphi) = \bar{v}(\psi).$$

Suppose not. Then $\bar{v}(\varphi) \neq \bar{v}(\psi)$ in B. By Corollary 1.3.18 ("Ultrafilters separate elements"), distinct elements of a Boolean algebra can be separated by a homomorphism into 2. So there is a Boolean homomorphism

$$h\colon B \longrightarrow 2$$

such that $h(\bar{v}(\varphi)) \neq h(\bar{v}(\psi))$.

$$\begin{tikzcd} \mathcal{L} \arrow[r] \arrow[dr, "v"'] & \mathrm{Form}(\mathcal{L}) \arrow[r, two heads, "\pi"] \arrow[d, "\bar{v}"] & \mathrm{Form}(\mathcal{L})/{\equiv} \arrow[dl, dashed, "\widetilde{v}"] \\ & B \arrow[d, "h"] & \\ & 2 & \end{tikzcd}$$

Consider now the composite

$$h \circ v\colon \mathcal{L} \longrightarrow 2.$$

This is a valuation on $\mathcal{L}$, and its unique extension to a homomorphism

$$\mathrm{Form}(\mathcal{L}) \longrightarrow 2$$

is precisely $h \circ \bar{v}$. So we have another commutative triangle:

$$\begin{tikzcd} \mathcal{L} \arrow[r, hook] \arrow[dr, "h \circ v"'] & \mathrm{Form}(\mathcal{L}) \arrow[d, "h \circ \bar{v}"] \\ & 2. \end{tikzcd}$$

Thus

$$(h \circ \bar{v})(\varphi) \neq (h \circ \bar{v})(\psi),$$

which means that the valuation $h \circ v$ distinguishes φ and ψ. But this contradicts the assumption that $\varphi \equiv \psi$.

Therefore $\bar{v}$ is constant on $\equiv$-equivalence classes, so it factors uniquely through $\mathrm{Form}(\mathcal{L})/\equiv$. Hence, $\mathrm{Form}(\mathcal{L})/\equiv$ is the free Boolean algebra on $\mathcal{L}$. ■

Proposition 1.7.20 (The Stone dual of the free Boolean algebra on $\mathcal{L}$)**.** *For every set $\mathcal{L}$, the Stone dual of the free Boolean algebra $\mathrm{Free}_{\mathsf{BA}}(\mathcal{L})$ on $\mathcal{L}$ is homeomorphic to $2^{\mathcal{L}}$ equipped with the product topology.*

Proof. Since the free Boolean algebra functor

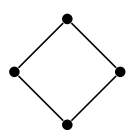
$$\mathrm{Free}_{\mathsf{BA}} \colon \mathsf{Set} \longrightarrow \mathsf{BA}$$

is left adjoint to the forgetful functor, it preserves coproducts. Hence

$$\mathrm{Free}_{\mathsf{BA}}(\mathcal{L}) \cong \mathrm{Free}_{\mathsf{BA}}\left(\sum_{p \in \mathcal{L}} 1\right) \cong \sum_{p \in \mathcal{L}} \mathrm{Free}_{\mathsf{BA}}(1),$$

where 1 denotes a singleton. By Stone duality, coproducts in BA correspond to products in Stone. Therefore the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(\mathcal{L})$ is homeomorphic to the product of $|\mathcal{L}|$ copies of the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(1)$.

It remains to identify the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(1)$. The free Boolean algebra on one generator is easily seen to be the four-element Boolean algebra,



and so it is isomorphic to $\mathcal{P}(2)$. Hence, its Stone dual is the two-point discrete space (which we denote by 2).[a]

Therefore, the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(\mathcal{L})$ is homeomorphic to $2^{\mathcal{L}}$ with the product topology. ■

[a] Alternatively, without explicitly identifying the free Boolean algebra on one generator, one can use the fact that $\mathrm{Free}_{\mathsf{BA}}$ is left adjoint to the forgetful functor, and hence the points of the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(1)$ are in bijection with

$$\hom_{\mathsf{BA}}(\mathrm{Free}_{\mathsf{BA}}(1), 2) \cong \hom_{\mathsf{Set}}(1, 2) \cong 2,$$

to obtain that the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(1)$ is the two-point discrete space.

Corollary 1.7.21 (How many formulas are there up to semantic equivalence?)**.** *If $\mathcal{L}$ is finite of cardinality n, then*

$$\mathrm{Form}(\mathcal{L})/\equiv \cong \mathcal{P}(2^{\mathcal{L}}),$$

and therefore

$$|\mathrm{Form}(\mathcal{L})/\equiv| = 2^{2^n}.$$

So, there are 2^{2^n} many formulas in n propositional symbols up to semantic equivalence.

Proof. If $|\mathcal{L}| = n < \infty$, then $2^{\mathcal{L}}$ is a finite discrete space. Hence, every subset of $2^{\mathcal{L}}$ is clopen, so

$$\mathrm{Clop}(2^{\mathcal{L}}) = \mathcal{P}(2^{\mathcal{L}}).$$

By Proposition 1.7.20 and Stone duality,

$$\mathrm{Form}(\mathcal{L})/{\equiv} \cong \mathrm{Clop}(2^{\mathcal{L}}) = \mathcal{P}(2^{\mathcal{L}}).$$

Since $|2^{\mathcal{L}}| = 2^n$, it follows that

$$|\mathrm{Form}(\mathcal{L})/{\equiv}| = |\mathcal{P}(2^{\mathcal{L}})| = 2^{2^n}.$$

■

Example 1.7.22. If $\mathcal{L} = \varnothing$, then there are 2 formulas up to equivalence ($[0], [1]$); if $\mathcal{L} = \{p\}$, then there are 4 ($[0], [1], [p], [\neg p]$); if $\mathcal{L} = \{p, q\}$, then there are 16.

1.7.9 Exercises on Stone duality

The following exercises are meant to reinforce the use of Stone duality. I would solve them (or get the intuition for them) via Stone duality (and that's the reason why they are here), as it is easier for me to think via Stone spaces, although for you it may be different. Try to discover your own preferred point of view.

Exercise 1.7.23. How many endomorphisms does an 8-element Boolean algebra have? How many automorphisms does it have?

Solution. Every Boolean algebra with 8 elements is isomorphic to $\mathcal{P}(X)$ with $|X| = 3$. Indeed, every finite Boolean algebra of cardinality 2^n is isomorphic to the powerset algebra of an n-element set, and here $8 = 2^3$.

By Stone duality, endomorphisms of $\mathcal{P}(X)$ correspond to continuous maps from the discrete Stone space X on 3 points to itself. Since every map between finite discrete spaces is continuous, the number of endomorphisms is just the number of functions from 3 to 3, which is 27.

Similarly, automorphisms of $\mathcal{P}(X)$ correspond to homeomorphisms between X and itself, i.e., to permutations of a 3-element set. Therefore, the number of automorphisms is $|S_3| = 3! = 6$. ■

Exercise 1.7.24. How many elements does the coproduct in BA of an 8-element Boolean algebra and a 16-element Boolean algebra have?

Solution. If B_1 has $8 = 2^3$ elements and B_2 has $16 = 2^4$ elements, then $B_1 \cong \mathcal{P}(X)$ and $B_2 \cong \mathcal{P}(Y)$ with $|X| = 3$ and $|Y| = 4$. By Stone duality,

$$B_1 + B_2 \;\cong\; \mathrm{Clop}(X \times Y) = \mathcal{P}(X \times Y),$$

and so

$$|B_1 + B_2| = |\mathcal{P}(X \times Y)| = 2^{|X \times Y|} = 2^{3 \cdot 4} = 2^{12}.$$

Therefore, the solution is $2^{12} = 4096$. ■

Exercise 1.7.25. Let A be an infinite Boolean algebra and B a finite nontrivial Boolean algebra. Show that B is isomorphic to a subalgebra of A.

Solution. Under Stone duality, this corresponds to the following:

Let X be an infinite Stone space and F a finite nonempty discrete space.
Show that there is a surjective continuous map from X to F.

It is enough to partition X into $n \coloneqq |F|$ pairwise disjoint nonempty clopen subsets.

We prove by induction on $n \geq 1$ that every infinite Stone space admits such a partition. The case $n = 1$ is trivial. Assume we have a partition

$$X = U_1 \sqcup \cdots \sqcup U_k$$

into k pairwise disjoint nonempty clopen subsets. Since X is infinite, at least one of the U_i's must be infinite; let it be U_j. Now U_j, being a clopen subspace of a Stone space, is again a Stone space. Choose two distinct points $x, y \in U_j$, which exist because U_j is infinite. Since U_j is totally separated, there is a clopen subset

$$V \subseteq U_j$$

such that $x \in V$ and $y \notin V$. Then both V and $U_j \setminus V$ are nonempty clopen subsets of the clopen U_j (and hence of X). Replacing U_j by these two pieces, we obtain a partition of X into $k+1$ pairwise disjoint nonempty clopen subsets. Thus, by induction, X admits a partition

$$X = U_1 \sqcup \cdots \sqcup U_n$$

into n pairwise disjoint nonempty clopen subsets.

Let $z_1, \ldots, z_n$ denote the elements of F. Define the function $q\colon X \twoheadrightarrow F$ by setting $q(x) = z_i$ whenever $x \in U_i$. It is surjective because each U_i is nonempty. Moreover, since F is discrete and each fiber $q^{-1}[\{z_i\}] = U_i$ is clopen, the map q is continuous. ■

The assertion in Exercise 1.7.25 can be strenghtened as follows.

Exercise 1.7.26. Let A be an infinite Boolean algebra and B a finite nontrivial Boolean algebra. Show that B is a retract of A, i.e., that there are Boolean homomorphisms $\iota\colon B \hookrightarrow A$ and $\pi\colon A \twoheadrightarrow B$ such that the composite $B \overset{\iota}{\hookrightarrow} A \overset{\pi}{\twoheadrightarrow} B$ is the identity on B.

(By $\pi \circ \iota = \mathrm{id}_B$, one necessarily has that ι is injective and π is surjective.)

Solution. Under Stone duality, this corresponds to the following:

Let X be an infinite Stone space and F a finite nonempty discrete space.
Show that there are continuous maps $q\colon X \twoheadrightarrow F$ and $j\colon F \hookrightarrow X$
such that the composite $F \overset{j}{\hookrightarrow} X \overset{q}{\twoheadrightarrow} F$ is the identity on F.

So, let X be an infinite Stone space and F a finite nonempty discrete space. By the argument in Exercise 1.7.25, there is a continuous surjective function $q\colon X \twoheadrightarrow F$. Since q is surjective, for every $z \in F$ we can choose a point $x_z \in q^{-1}[\{z\}]$. Define $j\colon F \hookrightarrow X$ by setting $j(z) \coloneqq x_z$ for every $z \in F$. Then, by construction, $q(j(z)) = z$ for every $z \in F$; i.e., the composite $F \overset{j}{\hookrightarrow} X \overset{q}{\twoheadrightarrow} F$ is the identity on F. Moreover, since F is discrete, every function from F to X is continuous, and hence j is continuous. ■

Exercise 1.7.27. Is the free Boolean algebra $\mathrm{Free}_{\mathsf{BA}}(\omega)$ on countably many generators complete?

Hint: you can use the following facts:

1. A Boolean algebra is complete if and only if its dual Stone space is *extremally disconnected* (i.e. the closure of every open set is clopen; equivalently, the closures of any two disjoint open sets are disjoint).
2. The Stone dual of $\mathrm{Free}_{\mathsf{BA}}(\omega)$ is homeomorphic to $2^{\mathbb{N}}$ with the product topology, which in turn is homeomorphic to the classical Cantor space $C \subseteq [0, 1]$.

Solution. No.

By the hint, it is enough to show that the Cantor space $C \subseteq [0, 1]$ is not extremally disconnected. To do so, we find two disjoint open subsets of the Cantor space whose closures intersect. Consider the open subsets of C

$$U \coloneqq C \cap (-\infty, 1/4), \qquad V \coloneqq C \cap (1/4, +\infty).$$

They are disjoint and open in the subspace topology on C. However, their closures in C intersect, since $1/4 \in \overline{U} \cap \overline{V}$. Therefore C is not extremally disconnected, and hence $\mathrm{Free}_{\mathsf{BA}}(\omega)$ is not complete.[a] ■

[a](Logical rephrasing: completeness would amount to saying that for any set Γ of formulas there is a single formula ψ such that for every formula φ one has $\Gamma \models \varphi$ if and only if $\psi \models \varphi$, which fails in classical propositional logic.)

Exercise 1.7.28. Let $\mathrm{FC}(\mathbb{N})$ be the algebra of finite and cofinite subsets of $\mathbb{N}$ (Example 1.2.15). Is the coproduct $\mathrm{FC}(\mathbb{N}) + \mathrm{FC}(\mathbb{N})$ isomorphic to $\mathrm{FC}(\mathbb{N})$?

Solution. No.

Let $X \coloneqq \mathrm{Ult}(\mathrm{FC}(\mathbb{N}))$. As discussed in Example 1.2.15, the Stone space X is homeomorphic to the one-point compactification $\mathbb{N} \cup \{\infty\}$ of the discrete space $\mathbb{N}$: the opens are the sets not containing ∞ and the cofinite sets containing ∞.

Now, by Stone duality, coproducts in BA correspond to products in Stone. Therefore, if one had $\mathrm{FC}(\mathbb{N})+\mathrm{FC}(\mathbb{N}) \cong \mathrm{FC}(\mathbb{N})$, then one would also have a homeomorphism $X \times X \cong X$.

We show that this is impossible. In X, there is exactly one non-isolated point, namely ∞. On the other hand, in $X \times X$, the isolated points are exactly the pairs (n, m) for $n, m \in \mathbb{N}$. Thus, X has exactly one non-isolated point—namely, ∞—, whereas $X \times X$ has infinitely many non-isolated points—namely, (∞, ∞), (n, ∞) for $n \in \mathbb{N}$, and (∞, n) for $n \in \mathbb{N}$. So, $X \times X$ is not homeomorphic to X. Therefore, $\mathrm{FC}(\mathbb{N}) + \mathrm{FC}(\mathbb{N})$ is not isomorphic to $\mathrm{FC}(\mathbb{N})$. ■

Remark 1.7.29. Every Boolean algebra embeds into a complete Boolean algebra. (This is simply by Stone's Representation Theorem.)

Chapter 2

Classical first-order logic

In the second part of the course, we move from classical propositional logic to classical first-order logic, where quantifiers enter the picture. As in the propositional case, we will relate *algebras of formulas* to *spaces of models* via a categorical duality.

2.1 From classical first-order logic towards Boolean doctrines

2.1.1 Syntax: first-order languages, formulas, free variables, sentences, substitutions

Fix once and for all an infinite supply of variables, typically denoted by

$$x, y, z, \dots$$

Definition 2.1.1 (Relational first-order language, relation symbol, arity)**.** A ***(one-sorted relational first-order) language*** $\mathcal{L}$ consists of a set, whose elements are called ***relation symbols***, together with an assigned natural number for each relation symbol. The number assigned to a relation symbol R is called the ***arity*** of R, and is denoted $\mathrm{ar}(R)$.
In this chapter, we consider only the relational one-sorted case, and we do not include equality.

Relation symbols are typically denoted by $R, S, \dots$

Example 2.1.2 (Graphs and preorders)**.** A language with one binary relation symbol can be used to describe directed graphs. Indeed, let $\mathcal{L} = \{R\}$, where R has arity 2. Then, we can form the expression

$$R(x, y),$$

which one may read informally as "there is an edge from x to y".

The same syntactic language can also be used to describe preorders. For this, one may write the unique binary relation symbol as $\leq$, so that the atomic formulas have the form

$$x \leq y.$$

The difference between graphs and preorders is not in the language, but in the axioms that we impose later on the structures interpreting that language.

Definition 2.1.3 (Formula)**.** The set of ***formulas*** $\mathrm{Form}(\mathcal{L})$ is defined inductively as follows:

- if $R \in \mathcal{L}$ has arity n, and $x_1, \dots, x_n$ are (not necessarily distinct) variables, then $R(x_1, \dots, x_n) \in$

Form($\mathcal{L}$) (the formulas of this form are called ***atomic formulas***);

- if $\varphi, \psi \in \mathrm{Form}(\mathcal{L})$, then $(\varphi \vee \psi) \in \mathrm{Form}(\mathcal{L})$ and $(\varphi \wedge \psi) \in \mathrm{Form}(\mathcal{L})$;
- if $\varphi \in \mathrm{Form}(\mathcal{L})$, then $(\neg\varphi) \in \mathrm{Form}(\mathcal{L})$;
- $0, 1 \in \mathrm{Form}(\mathcal{L})$;
- if x is a variable and $\varphi \in \mathrm{Form}(\mathcal{L})$, then $(\exists x\, \varphi) \in \mathrm{Form}(\mathcal{L})$ and $(\forall x\, \varphi) \in \mathrm{Form}(\mathcal{L})$.

In other words, Form($\mathcal{L}$) is the smallest set containing all atomic formulas and closed under the Boolean connectives and the quantifiers.

Formulas are typically denoted by $\varphi, \psi, \sigma, \ldots$.

Remark 2.1.4. In principle, quantifiers may be nested using the same variable symbol. For example, the expression

$$\forall x\, (R(x) \wedge \forall x\, P(x))$$

is a well-formed formula. However, it should be thought of as a needlessly confusing variant of

$$\forall x\, (R(x) \wedge \forall y\, P(y)).$$

Similarly, a formula of the form

$$R(x) \wedge \forall x\, P(x)$$

should be thought of as a needlessly confusing variant of

$$R(x) \wedge \forall y\, P(y).$$

Definition 2.1.5 (Free and bound occurrences, free variable, sentence)**.** Let $\mathcal{L}$ be a one-sorted relational first-order language, and let $\varphi \in \mathrm{Form}(\mathcal{L})$.
An occurrence of a variable x in φ is said to be ***bound*** if either it is the occurrence of x in a quantifier $\exists x$ or $\forall x$, or it lies within the scope of a quantifier $\exists x$ or $\forall x$. Otherwise, the occurrence is said to be ***free***.
A ***free variable*** of φ is a variable having a free occurrence in φ. We denote by

$$\mathrm{FV}(\varphi)$$

the set of free variables of φ.
A formula with no free variables is called a ***sentence***.

Example 2.1.6 (Free and bound occurrences)**.** In the formula

$$R(\underset{\substack{\uparrow \\ \text{free}}}{x}, y) \wedge \forall \underset{\substack{\uparrow \\ \text{bound}}}{x}\, S(\underset{\substack{\uparrow \\ \text{bound}}}{x}, z),$$

the first occurrence of x is free, while the second and the third occurrences of x are bound. Since the formula has a free occurrence of x, the variable x is a free variable of this formula.

Example 2.1.7 (Free variables)**.** Consider a language with one binary relation symbol R. Then:

$$\mathrm{FV}(R(x, y)) = \{x, y\},$$
$$\mathrm{FV}(\forall x\, R(x, y)) = \{y\},$$
$$\mathrm{FV}(\exists y\, \forall x\, R(x, y)) = \varnothing.$$

So $R(x, y)$ and $\forall x\, R(x, y)$ are not sentences, whereas $\exists y\, \forall x\, R(x, y)$ is a sentence.

Before defining the semantics of formulas, it is useful to isolate one technical point that will later play an important role: substitution.

Definition 2.1.8 (Substitution of a variable by a variable)**.** Let $\mathcal{L}$ be a relational first-order language, let $\varphi \in \mathrm{Form}(\mathcal{L})$, and let x, y be variables. The formula obtained by ***substituting*** y ***for*** x ***in*** φ is denoted by

$$\varphi[y/x].$$

It is obtained by replacing the *free* occurrences of x in φ by y. (Bound occurrences are not replaced.) Moreover, when passing through a quantifier, one must avoid ***variable capture***: if replacing a free occurrence of x by y would make it fall under a quantifier binding y, then one first renames that bound variable to a fresh variable.

For example, if one starts from the formula

$$\forall y\, R(x, y),$$

then the substitution of y for x is *not*

$$\forall y\, R(y, y),$$

because, in this formula, the new occurrence of y would become bound. Instead, one first renames the bound variable, for instance from y to z, and obtains

$$\forall z\, R(x, z),$$

so that

$$(\forall y\, R(x, y))[y/x] = \forall z\, R(y, z).$$

2.1.2 Semantics: structures, assignments, validity of formulas, models, semantic equivalence

We now pass from syntax to semantics.

Definition 2.1.9 (Structure)**.** Let $\mathcal{L}$ be a relational first-order language. An ***$\mathcal{L}$-structure*** is a set[a] A equipped, for each relation symbol $R \in \mathcal{L}$, with a subset (denoted by R^A and called the *interpretation of R in A*) of

$$\underbrace{A \times \cdots \times A}_{\mathrm{ar}(R)\text{ times}}.$$

In other words, each relation symbol R is interpreted in A as an $(\mathrm{ar}(R))$-ary relation on the set A.

[a] Unlike many standard treatments of first-order logic, we allow the underlying set of an $\mathcal{L}$-structure to be empty.

For example, if $\mathcal{L} = \{\leq\}$ and $\mathrm{ar}(\leq) = 2$, then an $\mathcal{L}$-structure is just a set equipped with a binary relation.

Remark 2.1.10 (Nullary relation symbols)**.** A relation symbol of arity 0 behaves like a propositional symbol. Indeed, if $R \in \mathcal{L}$ and $\mathrm{ar}(R) = 0$, then in every $\mathcal{L}$-structure A its interpretation is a subset

$$R^A \subseteq A^0.$$

But A^0 is a singleton, namely the set consisting of the unique function from the empty set to A. Therefore, R^A is either empty or all of A^0.

So one may think of the interpretation of R as simply *false* in the first case, and *true* in the second. In this sense, nullary relation symbols play the role of the propositional symbols of classical propositional logic.

Definition 2.1.11 (Assignment)**.** Let A be an $\mathcal{L}$-structure, and let X be a finite set of variables. An ***assignment on*** X ***in*** A is a function

$$\nu \colon X \to A.$$

Definition 2.1.12 (Validity of a formula in a structure under an assignment ($\models$))**.** Let $\mathcal{L}$ be a relational first-order language and let A be an $\mathcal{L}$-structure. By induction on the complexity of a formula φ in the language $\mathcal{L}$, we define, for every finite set X of variables with $\mathrm{FV}(\varphi) \subseteq X$ and every function $\nu\colon X \to A$, the expression

$$A, \nu \models \varphi,$$

read as "φ holds in A under the assignment ν", as follows.

- $A, \nu \models R(x_1, \dots, x_n) \iff (\nu(x_1), \dots, \nu(x_n)) \in R^A$, for $R \in \mathcal{L}$ of arity n and $x_1, \dots, x_n \in X$;
- $A, \nu \models \varphi \vee \psi \iff A, \nu \models \varphi$ or $A, \nu \models \psi$;
- $A, \nu \models \varphi \wedge \psi \iff A, \nu \models \varphi$ and $A, \nu \models \psi$;
- $A, \nu \models \neg\varphi \iff$ not $A, \nu \models \varphi$;
- $A, \nu \models 0 \iff$ false (i.e., $A, \nu \models 0$ never holds);
- $A, \nu \models 1 \iff$ true (i.e., $A, \nu \models 1$ always holds);
- $A, \nu \models \exists x\, \varphi$ if and only if, picking any variable[a] $z \notin X$ (such as x itself if $x \notin X$), there is $a \in A$ such that
$$A, \nu' \models \varphi[z/x],$$
where
$$\nu'\colon X \cup \{z\} \to A$$
is given by
$$\nu'(y) = \begin{cases} \nu(y) & \text{if } y \in X, \\ a & \text{if } y = z; \end{cases}$$
- $A, \nu \models \forall x\, \varphi$ if and only if, picking any variable $z \notin X$ (such as x itself if $x \notin X$), for every $a \in A$ we have
$$A, \nu' \models \varphi[z/x],$$
where
$$\nu'\colon X \cup \{z\} \to A$$
is given by
$$\nu'(y) = \begin{cases} \nu(y) & \text{if } y \in X, \\ a & \text{if } y = z. \end{cases}$$

[a] One can prove that the ensuing condition is independent of the choice of the variable z (as well as of any possible additional choice of variable needed to avoid variable capture in the substitution $\varphi[z/x]$). The same holds for the case $\forall$.

Definition 2.1.13 (Validity of a sentence in a structure ($\models$))**.** Let $\mathcal{L}$ be a relational first-order language, let A be an $\mathcal{L}$-structure, and let φ be a sentence in the language $\mathcal{L}$. We write

$$A \models \varphi$$

if

$$A, \varnothing \models \varphi,$$

where $\varnothing\colon \varnothing \to A$ denotes the unique function from the empty set to A.

Example 2.1.14. Let $\mathcal{L} = \{R\}$, where R is a binary relation symbol, and let A be an $\mathcal{L}$-structure. We want

to understand what the sentence

$$\exists x\, \forall y\, R(x, y)$$

means in A.

By definition,

$$A \models \exists x\, \forall y\, R(x, y)$$

means

$$A, \varnothing \models \exists x\, \forall y\, R(x, y), \tag{2.1}$$

where $\varnothing \colon \varnothing \to A$ is the unique function from the empty set to A.

Now pick a variable z not in the empty set. By the case $\exists$ in the definition of $\models$ (Definition 2.1.12), the condition (2.1) means that there is $a \in A$ such that

$$A, \nu \models \forall y\, R(z, y),$$

where

$$\nu \colon \{z\} \to A$$

is given by $\nu(z) = a$.

Next, by the case $\forall$ in the definition of $\models$ (Definition 2.1.12), this means that, picking a variable $w \notin \{z\}$, for every $b \in A$ we have

$$A, \nu' \models R(z, w),$$

where

$$\nu' \colon \{z, w\} \to A$$

is given by

$$\nu'(z) = a, \qquad \nu'(w) = b.$$

By the definition of validity for atomic formulas, this is equivalent to saying that

$$(a, b) \in R^A$$

for every $b \in A$.

Therefore,

$$A \models \exists x\, \forall y\, R(x, y)$$

holds if and only if there is $a \in A$ such that for every $b \in A$ one has $(a, b) \in R^A$.

We can now incorporate semantic assumptions, just as we did in the propositional case.

Definition 2.1.15 (Theory)**.** Let $\mathcal{L}$ be a relational first-order language. A ***theory*** in the language $\mathcal{L}$ is a set of sentences in the language $\mathcal{L}$.

Definition 2.1.16 (Model of a theory)**.** Let $\mathcal{T}$ be a theory in a relational first-order language $\mathcal{L}$. A ***model*** of $\mathcal{T}$ is an $\mathcal{L}$-structure A such that, for every sentence $\varphi \in \mathcal{T}$, $A \models \varphi$.

Definition 2.1.17 (Semantic equivalence modulo a theory, in a context)**.** Let $\mathcal{T}$ be a theory in the language $\mathcal{L}$, let X be a finite set of variables, and let $\varphi, \psi \in \mathrm{Form}(\mathcal{L})$ be such that

$$\mathrm{FV}(\varphi) \subseteq X \qquad \text{and} \qquad \mathrm{FV}(\psi) \subseteq X.$$

We say that φ and ψ are ***semantically equivalent modulo $\mathcal{T}$ in the context*** X if, for every model A of $\mathcal{T}$ and every assignment

$$\nu \colon X \to A,$$

one has

$$A, \nu \models \varphi \qquad \Longleftrightarrow \qquad A, \nu \models \psi.$$

In other words, once the assumptions in $\mathcal{T}$ are fixed, two formulas are identified in the context X when they have the same truth value in every model of $\mathcal{T}$ under every assignment of values to the variables in X.

The next example shows a case in which two sentences are not equivalent in the empty context, but become equivalent in any nonempty context.

Example 2.1.18 (Dependence on the context)**.** Let $\mathcal{L}$ be the empty language, let $\mathcal{T}$ be the empty theory, and consider the formula

$$\exists x\, 1.$$

We first look at the empty context $\varnothing$. In this context, the formula $\exists x\, 1$ is *not* semantically equivalent modulo $\mathcal{T}$ to 1. Indeed, if $A = \varnothing$, then A is an $\mathcal{L}$-structure and hence a model of $\mathcal{T}$. Now

$$A \models 1,$$

but

$$A \not\models \exists x\, 1,$$

since there is no element of A that can witness the existential quantifier.

We now look at the context $\{y\}$. In this context, the formula $\exists x\, 1$ *is* semantically equivalent modulo $\mathcal{T}$ to 1. Indeed, let A be any model of $\mathcal{T}$, and let

$$\nu \colon \{y\} \to A$$

be any assignment. Then A must be non-empty, since otherwise no such assignment could exist. Hence there exists some $a \in A$, and therefore

$$A, \nu \models \exists x\, 1.$$

On the other hand,

$$A, \nu \models 1$$

always holds. So, for every model A of $\mathcal{T}$ and every assignment $\nu \colon \{y\} \to A$, we have

$$A, \nu \models \exists x\, 1 \qquad \Longleftrightarrow \qquad A, \nu \models 1.$$

Thus $\exists x\, 1$ is not equivalent to 1 in the empty context, but it is equivalent to 1 in the context $\{y\}$. A similar reasoning shows that $\exists x\, 1$ is equivalent to 1 in any nonempty context.

The point is that choosing a context amounts to declaring which variables are available as *generic elements* of the structure. If the context is non-empty, then only non-empty structures admit assignments on that context.

2.2 First-order Boolean doctrines

The previous discussion suggests that, in first-order logic, one should not look at a single Boolean algebra of formulas, but rather at a family of Boolean algebras: one for each finite set of variables. Here we only model one-sorted purely relational first-order logic without equality.

Let FinSet denote the category of finite sets and functions, and recall that BA is the category of Boolean algebras and Boolean homomorphisms. The following notion—of first-order Boolean doctrine—is essentially due to Lawvere ([Lawvere, 1970])[1]. To understand the definition, I suggest comparing each passage of the definition with its meaning in the intended example put right after: the Lindenbaum–Tarski first-order Boolean doctrine of a theory (Definition 2.2.2).

[1] Lawvere introduced *hyperdoctrines*, whose definition is similar, although not identical, to the one of first-order Boolean doctrines that we give here.

Definition 2.2.1 (First-order Boolean doctrine)**.** A ***first-order Boolean doctrine*** is a functor

$$P\colon \mathsf{FinSet} \to \mathsf{BA}$$

with the following properties.

1. (Quantifiers as adjoints to adding a dummy variable) For every finite set X and every $y \notin X$, the Boolean homomorphism[a]

$$P(X \hookrightarrow X \cup \{y\})\colon P(X) \longrightarrow P(X \cup \{y\})$$

has both a left adjoint and a right adjoint (as order-preserving functions between posets), denoted respectively by

$$(\exists y)_X\colon P(X \cup \{y\}) \longrightarrow P(X) \qquad \text{and} \qquad (\forall y)_X\colon P(X \cup \{y\}) \longrightarrow P(X).$$

This means that, for every $\varphi \in P(X \cup \{y\})$, there are (necessarily unique) elements $(\exists y)_X\, \varphi$ and $(\forall y)_X\, \varphi$ such that, for every $\psi \in P(X)$,

$$(\exists y)_X\, \varphi \leq \psi \iff \varphi \leq P(X \hookrightarrow X \cup \{y\})(\psi) \qquad \text{and} \qquad P(X \hookrightarrow X \cup \{y\})(\psi) \leq \varphi \iff \psi \leq (\forall y)_X\, \varphi.$$

2. (Beck–Chevalley property: substitutions commute with quantifiers) For every function $\sigma\colon X \to X'$ between finite sets, every $y \notin X$ and every $y' \notin X'$, calling $\widetilde{\sigma}\colon X \cup \{y\} \to X' \cup \{y'\}$ the extension of σ that maps y to y', the following diagrams commute.

$$\begin{tikzcd} P(X \cup \{y\}) \arrow[r, "(\exists y)_X"] \arrow[d, "P(\widetilde{\sigma})"'] & P(X) \arrow[d, "P(\sigma)"] \\ P(X' \cup \{y'\}) \arrow[r, "(\exists y')_{X'}"'] & P(X') \end{tikzcd} \qquad \begin{tikzcd} P(X \cup \{y\}) \arrow[r, "(\forall y)_X"] \arrow[d, "P(\widetilde{\sigma})"'] & P(X) \arrow[d, "P(\sigma)"] \\ P(X' \cup \{y'\}) \arrow[r, "(\forall y')_{X'}"'] & P(X') \end{tikzcd}$$

[a] Intuitively, the Boolean homomorphism $P(X \hookrightarrow X \cup \{y\})\colon P(X) \to P(X \cup \{y\})$ corresponds to adding a dummy variable.

Definition 2.2.2 (The Lindenbaum–Tarski first-order Boolean doctrine)**.** Let $\mathcal{T}$ be a theory in a one-sorted relational first-order language $\mathcal{L}$, with no equality. The ***Lindenbaum–Tarski first-order Boolean doctrine of*** $\mathcal{T}$ is the functor

$$\mathrm{LT}_{\mathcal{T}}\colon \mathsf{FinSet} \to \mathsf{BA}$$

defined as follows.
For every finite set X of variables, the Boolean algebra $\mathrm{LT}_{\mathcal{T}}(X)$ is the quotient of the set of formulas $\varphi \in \mathrm{Form}(\mathcal{L})$ such that

$$\mathrm{FV}(\varphi) \subseteq X$$

by semantic equivalence modulo $\mathcal{T}$ in the context X.
For every function $\sigma\colon X \to Y$ of finite sets, the Boolean homomorphism

$$\mathrm{LT}_{\mathcal{T}}(\sigma)\colon \mathrm{LT}_{\mathcal{T}}(X) \to \mathrm{LT}_{\mathcal{T}}(Y)$$

is induced by simultaneous substitution along σ: it sends the equivalence class of a formula φ to the equivalence class of the formula obtained by replacing each free occurrence of a variable $x \in X$ by the variable $\sigma(x) \in Y$, renaming bound variables when necessary in order to avoid variable capture.

Finally, for every finite set X and every variable $y \notin X$, the adjoints

$$(\exists y)_X, (\forall y)_X \colon \mathrm{LT}_{\mathcal{T}}(X \cup \{y\}) \to \mathrm{LT}_{\mathcal{T}}(X)$$

are induced by the usual existential and universal quantification:

$$(\exists y)_X[\varphi] = [\exists y\, \varphi], \qquad (\forall y)_X[\varphi] = [\forall y\, \varphi].$$

One checks that $\mathrm{LT}_{\mathcal{T}}$ is indeed a first-order Boolean doctrine.

Remark 2.2.3. The use of FinSet reflects the fact that, in this chapter, we are only considering <u>one-sorted purely relational</u> first-order logic. If you are curious about how to include function symbols and multiple sorts, read the footnote[2]. If one wants to consider equality, one requires a further condition (sometimes called *elementarity*) that we do not mention here.

[2]To simultaneously allow function symbols and multiple sorts, an elegant way is to use *algebraic theories*, i.e. categories with finite products (no need for them to be small, here). A *first-order Boolean doctrine over a category* C *with finite products* is a functor $\mathsf{C} \to \mathsf{BA}$ satisfying certain conditions. The definition of first-order Boolean doctrine that we saw in Definition 2.2.1 is the special case $\mathsf{C} = \mathsf{FinSet}^{\mathrm{op}}$. Roughly speaking, the category C plays the role of the algebraic theory of the language under consideration.

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